SSB PERSONAL INTERVIEW PREPARATION HANDBOOK

PIQ-linked questions, certificate deep dives, factual answers and cross-question chains

Prepared for Shyamendra Singh
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Research cut-off: 16 October 2026
Personal source set: PIQ Online.pdf, 4 pages; Certificates .pdf, 25 pages; your current-role description.

Your combination of agricultural engineering, algorithm development, mountaineering, running and community service creates several connected interview routes. This handbook separates what only you can answer from knowledge you can study, without inventing experiences or treating participation as a higher achievement.

Scope and honesty: This is an interviewing-officer-style preparation resource, not an official SSB publication. I am not a serving or former interviewing officer and do not claim ten years of service. Questions are plausible practice prompts, not predictions or leaked questions. No finite question bank can guarantee that every possible question is covered.

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Two categories throughout: P = personalised answer required, with preparation suggestions and supporting facts. F = factual knowledge, with an answer and cross-question support. For mixed questions, study the F material but supply your own judgement and experience.

Reading key: Bracketed source keys link each factual card to the source register at the end. Worked examples are explicitly illustrative, not claims about your company or results. Coaching suggestions are recommendations, not official selection rules.

Navigation and preparation method

1. Evidence map and PIQ consistency audit

2. Personal-answer question bank: identity, family, education, motivation, work, leadership, service, hobbies

3. Factual knowledge: Agra and region; institutions; employer; engineering; autonomy; mathematics; mountaineering; sport; service

4. Current affairs as of 16 October 2026, with profile-linked cross-questions

5. Pressure chains, rapid-fire practice, revision sheets and sources

Use each question in three passes. First answer aloud without notes. Then check the factual support. Finally ask the cross-questions in a different order. Personal answers should contain a real situation, your exact action, the outcome and what changed afterward. Do not memorise a fabricated model personality.

Depth ladder. Prepare a 30-second explanation, a 90-second explanation and a technical whiteboard explanation for your job and each major certificate. These are practice formats, not prescribed SSB time limits.

Evidence boundary. A certificate establishes the printed participation, dates, grade or result. It does not establish every task you performed, a summit reached, a project deployed, a patent granted, or national-level selection. Unclear handwriting and apparent inconsistencies are marked for your verification.

Interview conduct suggestion. Answer the actual question first. Explain an unfamiliar technical term once. If unsure, say what you know and what you would verify. Correct a factual slip promptly. Do not criticise classmates, colleagues or previous assessors to strengthen your own account.

1 | Evidence map: every attached certificate

Certificate 01 | Virtual Labs internship appreciation

DEI Engineering Graphics Virtual Lab; MoE-NMEICT Virtual Labs III (Extended). Listed engagements from July 2023 to March 2026, with a gap in the listed periods during 2025. Earlier remuneration Rs 7,500/month; last listed period Rs 15,000/month.

Interview route: Explain exact modules, hours, deliverables, testing, stipend change and how work fitted around study.

Certificate 02 | Training and placement coordination

B.Tech Agricultural Engineering, institute attendance 2022-2026; student coordinator team involvement 2023-2026; certificate dated 6 October 2026.

Interview route: Separate coordination from authority to select candidates. Prepare one scheduling problem and your response.

Certificate 03 | NSS

2022-23 and 2023-24; 240 hours; special camp 18-24 December 2022.

Interview route: Recall adopted community, duties, programme officer, attendance records and actual medical-camp role.

Certificate 04 | NIMAS Basic Mountaineering Course

23 May-19 June 2026; grade A; recommendation for advanced mountaineering course.

Interview route: Do not convert recommendation into completion of the advanced course or a claim of summiting 6,500 m.

Certificate 05 | IMF sport climbing Level 1

Indian Mountaineering Foundation, New Delhi; grade A; handwritten date appears 25 July 2026.

Interview route: Confirm the handwritten date against your records; prepare techniques actually taught.

Certificate 06 | Faridabad Half Marathon 2026

Event certificate; printed distance 21 km; net time 02:17:01.

Interview route: Use printed distance for calculations; distinguish net time from gun time.

Certificate 07 | Bordermen Marathon 2026

BSF event, Amritsar to Attari; 22 February 2026; your category 10 km; chip time 00:53:32; printed position 287.

Interview route: The event name does not mean you ran a full marathon. Rank denominator/category is not shown.

Certificate 08 | Escorts Kubota practical training

7 April-25 August 2025; RNIL R&D, Faridabad; project on simulation of a self-guided tractor with control system using ROS.

Interview route: Explain simulator versus physical deployment and your individual contribution.

Certificate 09 | ICAR-IARI internship

1-30 June 2024; Agricultural Engineering; additive manufacturing and mechatronics for efficient farm machinery design.

Interview route: Prepare one manufactured component, process, material and validation method that you actually used.

Certificate 10 | Essay competition

IE(I) student chapters; session 2024-25; third-year Agricultural Engineering; handwritten result appears first position; verify against the original.

Interview route: PIQ lists an essay second position in 2021. This later certificate may be a separate event; confirm both records.

Certificate 11 | Quiz competition

IE(I) student chapters; session 2024-25; third-year Agricultural Engineering; first position.

Interview route: Recall format, team size, rounds and one difficult question without inventing it.

Certificate 12 | IPR workshop

One-day workshop on intellectual property rights/copyrights; 3 February 2024.

Interview route: Attendance is not a patent or copyright registration.

Certificate 13 | SIH internal hackathon 2024

Team Agri_Catalysts; internal hackathon 10 September 2024; certificate issued 22 September 2024.

Interview route: Internal participation is not national finalist status. Confirm exact problem statement and selection outcome.

Certificate 14 | SIH internal hackathon 2023

Team SRAJAN; internal hackathon 23 September 2023.

Interview route: Explain whether this is the same project as 2024 or a separate attempt.

Certificate 15 | fNIRS workshop assistance

Workshop on brain research methods using functional near-infrared spectroscopy, 3-7 March 2025; certificate of assistance.

Interview route: Describe logistics or technical help actually provided; do not claim neuroscience research expertise.

Certificate 16 | Blood donation appreciation

Dated 15 October 2025; donation-camp certificate.

Interview route: Prepare motivation and logistics. No need to disclose medical details or memorise eligibility rules without official checking.

Certificate 17 | School academic appreciation

Amar Ujala recognition for academic session 2018-19.

Interview route: Explain actual marks and scope of recognition, not a district rank unless documented.

Certificate 18 | School GK competition

Class 10; first position; 28 January 2019.

Interview route: Separate school-level GK from faculty-level quiz results.

Certificate 19 | Bhartiya Sanskriti Gyan Pariksha

2018-19, class 10; certificate shows category alternatives, but selected category/score is not reliably clear.

Interview route: Confirm result category from original. Discuss inclusive civic values rather than claiming an unreadable distinction.

Certificate 20 | RS-CIT

VMOU Kota; examination February 2017; 63%; name printed Shyamendra Tomar, father Kamlendra Tomar.

Interview route: Check name consistency and supporting records. Do not invent a legal explanation.

Certificate 21 | AI for Everyone module

DEI, session 2022; grade point 8.500; deemed percentage 85.000; first with distinction.

Interview route: Keep module percentage separate from B.Tech aggregate.

Certificate 22 | IIRS course merit

Basics of Remote Sensing, GIS and GNSS; 27 August-22 November 2024; 52 hours; online exam grade B; attendance 73%.

Interview route: Attendance percentage is not examination marks.

Certificate 23 | IIRS participation

Applications of Machine Learning in Urban Studies; 5-9 June 2023; 7 hours 30 minutes.

Interview route: Participation is not a graded qualification.

Certificate 24 | IIRS workshop

Remote sensing-based data analytics in agriculture; one-day online workshop, 26 October 2023.

Interview route: Prepare one agriculture use case and limitations of remote inference.

Certificate 25 | Intra-faculty quiz

Faculty of Science certificate; session appears 2021-22; second position; handwritten programme appears B.Sc.

Interview route: Verify programme and session. It may explain the 2021-22 timeline, but the document alone does not establish enrolment history.

Source for the complete inventory: [CERT], PDF pages 1-25. Handwriting is treated cautiously. Certificate praise is not used to evaluate you or other employees.

2 | PIQ consistency audit and missing personal facts

Education chronology

What the documents show: PIQ: class 12 in 2021 and B.Tech completion in 2026. Certificate 02: B.Tech attendance 2022-2026. Certificate 25 appears to refer to B.Sc./2021-22.

Prepare: What exactly did you do between school and engineering? Did you enrol in another programme? Prepare the truthful month-by-month chronology and records; do not label it a gap or a completed degree without evidence.

Academic percentages

What the documents show: PIQ reads 86.50% in matriculation, 80.20% in 10+2 and 80.51% in B.Tech.

Prepare: Check original marksheets, aggregation method and whether the final figure is provisional or final. Explain the fifth-semester distinction separately.

Essay result

What the documents show: PIQ lists second position in 2021; certificate 10 appears to show first in a later session. These may be separate events, not a contradiction.

Prepare: Different events may explain this. Match year, organiser, level and result; correct an actual entry error through the authorised process if applicable.

Hackathon years

What the documents show: PIQ mentions 2023; certificates cover internal events in 2023 and 2024.

Prepare: Prepare separate team names, problems, roles and outcomes. Do not merge the events.

Current residence population

What the documents show: PIQ gives about 1,590,000 but does not clearly identify present city in the visible fields.

Prepare: Do not infer present residence from your employer. Confirm city, municipal boundary and census/estimate year. Village, city, urban agglomeration and district populations differ.

Internship chronology

What the documents show: Virtual Labs listed engagements overlap academic years; IARI June 2024 and EKL April-August 2025 are also documented.

Prepare: Explain hours, remote/on-site arrangements, leave and deliverables. An overlap is a question to explain, not proof of a problem.

Name variation

What the documents show: RS-CIT uses Tomar while PIQ and most certificates use Singh.

Prepare: Check identity and education records. Explain only the actual reason and available documentation.

Role and remuneration

What the documents show: PIQ: R&D Engineer, Rs 36,000 monthly; previous project intern, Rs 15,000. Current employer/team comes from your message.

Prepare: Confirm joining date, gross versus take-home amount, employment type and exact designation. Certificate 08 supports training in 2025, not the current joining date.

Responsibility wording

What the documents show: PIQ responsibility field is visibly truncated after class captain and training/placement wording.

Prepare: Prepare exact title, tenure, number of people coordinated and limits of authority. Do not upgrade coordinator to placement head.

Sports level and attempts

What the documents show: PIQ shows school cricket and company-department volleyball. Previous SSB/commission attempts are not unambiguously populated.

Prepare: Confirm exact participation years, position and all actual attempts. Do not infer first attempt or NCC training from a blank/radio-button extraction.

Course/entry identification

What the documents show: PIQ displays SSC(NT) Men-125 and UPSC; selection centre Allahabad.

Prepare: Use the exact current call-up letter and notification for entry, course and reporting instructions. Do not substitute NCC-125 search results for your UPSC entry.

Personal facts not established by the attachments: current joining date; exact current residential city; family business type; sibling course; school/college commute; final-year project title; DIY project inventory; running training log; highest altitude actually reached; exact climbing route grades; hackathon problem IDs; number of recruits coordinated; previous SSB outcomes. Fill these from your own records, not from this handbook.

Sources: [PIQ], [CERT]. No claim here is an adverse finding; these are preparation checks.

3 | Category P: personal background and family

P001 | Introduce yourself without reading your PIQ.

Your answer preparation: Build a short factual sequence: Sarendhi upbringing, schooling, Agricultural Engineering, current R&D/path-planning work, and two activities you sustain. Include one current responsibility, not a list of adjectives.

Supporting knowledge: Your PIQ lists Sarendhi in Agra district and Hindi as mother tongue. Your current employer and algorithm role are from your message.

Cross-questions and support: Why should we remember you? What is not in your PIQ? Which activity receives most time?

Sources: [PIQ], [CERT].

P002 | What responsibilities do you handle at home?

Your answer preparation: List actual recurring tasks, financial contributions if any, and a recent decision you helped with. Distinguish helping from being solely responsible.

Supporting knowledge: The PIQ records a self-employed father, homemaker mother and younger brother studying; it does not identify the business or sibling course.

Cross-questions and support: How often do you visit? What happens when you are away? What did you do last month?

Sources: [PIQ], [CERT].

P003 | Describe your father’s work and what you understand about it.

Your answer preparation: Prepare business/service type, customer base, daily work, seasonal risks and how you have helped. Use approximate figures only when you know their basis.

Supporting knowledge: The PIQ gives monthly income Rs 15,000. Do not infer turnover, debt, assets or profitability from it.

Cross-questions and support: What is revenue versus income? How would you improve one process? What happens during a slow month?

Sources: [PIQ], [CERT].

P004 | What have you learned from your mother’s responsibilities?

Your answer preparation: Use a real household example involving planning, budgeting, care or resolving a practical issue. Describe your own contribution.

Supporting knowledge: Do not treat homemaking as absence of work or infer educational ability from a qualification.

Cross-questions and support: What task can you take over? How is the household budget decided?

Sources: [PIQ], [CERT].

P005 | What do you know about your younger brother’s education?

Your answer preparation: Prepare actual programme, institution, interests, fees if known, and how you support without making decisions for him.

Supporting knowledge: The PIQ only establishes undergraduate/student wording; course and year are not confirmed.

Cross-questions and support: What advice did you give recently? What if he chooses a different career?

Sources: [PIQ], [CERT].

P006 | How do you use your salary?

Your answer preparation: Prepare your real monthly budget: accommodation, food, commute, family support, savings, learning and recreation. Check that the total matches the correct take-home figure.

Supporting knowledge: PIQ income is Rs 36,000/month; gross and net are not distinguished.

Cross-questions and support: What is your largest expense? Could you manage an unexpected expense? Why spend on courses?

Sources: [PIQ], [CERT].

P007 | Describe a disagreement at home and your response.

Your answer preparation: Choose a real, ordinary disagreement. Explain the issue, what you said, what you changed and the outcome without attributing hidden motives to anyone.

Supporting knowledge: No personal disagreement is documented; do not invent one.

Cross-questions and support: What was your mistake? Did you compromise? What remains unresolved?

Sources: [PIQ], [CERT].

P008 | Who are your close friends and what do you do together?

Your answer preparation: Prepare factual duration of friendship, shared activities and an example of mutual help. Avoid ranking friends or assessing their personalities.

Supporting knowledge: Names and friendship history are not supplied.

Cross-questions and support: When did you last meet? What do you contribute? What disagreement did you resolve?

Sources: [PIQ], [CERT].

P009 | What is your daily routine on a working day and a holiday?

Your answer preparation: Use actual wake-up, commute, work, exercise, learning and sleep times. Explain variability during tests or travel.

Supporting knowledge: Cross-check against training logs and work hours rather than an idealised timetable.

Cross-questions and support: Where is the time for interview preparation? What do you stop doing when busy?

Sources: [PIQ], [CERT].

P010 | What is one habit you are improving?

Your answer preparation: Pick a real behaviour, its cost, a concrete corrective action and evidence of progress. Avoid a disguised strength.

Supporting knowledge: No weakness is inferred from your marks or certificates.

Cross-questions and support: When did it last cause a problem? What is your next measurable step?

Sources: [PIQ], [CERT].

P011 | What did you do in 2021-22?

Your answer preparation: Prepare exact admission, study, preparation, work or other activities with months. If a programme changed, explain the decision and outcome plainly.

Supporting knowledge: Certificate 25 appears to reference B.Sc. and 2021-22; confirm before describing it as enrolment.

Cross-questions and support: Why did you change? What did you complete? What would you do differently?

Sources: [PIQ], [CERT].

P012 | Tell me about an important setback.

Your answer preparation: Use a genuine academic, project, sport or administrative setback. Explain your responsibility, response and subsequent change.

Supporting knowledge: A certificate of success does not supply the setback narrative.

Cross-questions and support: What was within your control? Did you ask for help? Has the same issue recurred?

Sources: [PIQ], [CERT].

4 | Category P: education, choices and communication

P013 | Why Agricultural Engineering rather than another branch?

Your answer preparation: Explain the decision at the time: subjects, exposure, alternatives and constraints. Connect it to present work without rewriting history.

Supporting knowledge: Agricultural engineering spans machinery, processing, water and energy systems; see factual cards and [IARI].

Cross-questions and support: Was it your first preference? What would you choose now? Why not agriculture science?

Sources: [PIQ], [CERT].

P014 | Why did you choose Dayalbagh Educational Institute?

Your answer preparation: Prepare admission route, options, cost, commute and academic fit. Identify one feature you actually used.

Supporting knowledge: DEI’s official history places deemed-university status in 1981; do not confuse this with the origin of education at Dayalbagh.

Cross-questions and support: What did the institute change in your habits? What would you improve?

Sources: [PIQ], [CERT].

P015 | Explain the change from Hindi-medium schooling to English-medium engineering.

Your answer preparation: Give concrete methods: technical reading, speaking practice, terminology notes and project communication. Acknowledge remaining areas of practice.

Supporting knowledge: PIQ lists Hindi medium at school and English in engineering.

Cross-questions and support: Explain your project in Hindi now. Which technical term is hardest to translate?

Sources: [PIQ], [CERT].

P016 | Why did your marks change across school and college?

Your answer preparation: Use subject-wise evidence, workload and changes in study method. Avoid blaming teachers or making unsupported comparisons with peers.

Supporting knowledge: PIQ percentages: 86.50, 80.20 and 80.51; verify against originals.

Cross-questions and support: Which subject improved? Which examination was difficult? How did you prepare for semester V?

Sources: [PIQ], [CERT].

P017 | Which subject can you explain best?

Your answer preparation: Choose one genuinely understood subject. Prepare definitions, a derivation, an application and a limitation.

Supporting knowledge: For your role, geometry, optimisation, control and farm machinery are natural study routes, not assumed favourite subjects.

Cross-questions and support: Teach it to a non-engineer. Where have you used it? What is one common misconception?

Sources: [PIQ], [CERT].

P018 | What was your final-year project?

Your answer preparation: Prepare exact title, problem, input/output, your contribution, validation, limitations and current status. Do not substitute the internship project unless it was also your degree project.

Supporting knowledge: Final-year project title is not established in the attachments.

Cross-questions and support: What did you build yourself? What failed? How did you measure improvement?

Sources: [PIQ], [CERT].

P019 | What does highest marks in semester V mean?

Your answer preparation: State exact semester, marks, scope of comparison and documentary basis. Keep a semester distinction separate from overall university rank.

Supporting knowledge: PIQ claims highest marks in V semester, not overall university topper.

Cross-questions and support: Were you first overall? Which subject helped most? What was the next semester result?

Sources: [PIQ], [CERT].

P020 | Why is an early Faculty of Science quiz certificate in your file?

Your answer preparation: Check programme, session and event details on certificate 25. Explain the actual educational chronology and participation.

Supporting knowledge: The handwritten programme appears B.Sc.; this is not proof of degree completion.

Cross-questions and support: Were you in B.Sc.? Did you transfer credits? Why engineering later?

Sources: [PIQ], [CERT].

P021 | What did you write in the essay competition?

Your answer preparation: Recall actual title, central argument, two supporting points and your conclusion. Prepare a short opposing argument.

Supporting knowledge: Certificate 10 and PIQ appear to differ; verify whether separate competitions explain this.

Cross-questions and support: Was the result first or second? What would you change today?

Sources: [PIQ], [CERT].

P022 | What did you learn from the quiz competitions?

Your answer preparation: Prepare formats, dates, team or individual status and how you study beyond memorising facts.

Supporting knowledge: School GK first place, faculty quiz first place and earlier science-faculty second place are distinct records.

Cross-questions and support: Give a question you missed. How do you check a disputed answer?

Sources: [PIQ], [CERT].

5 | Category P: Army motivation, entry and service realism

P023 | Why join the Army when you already have an engineering career?

Your answer preparation: Explain the responsibilities you seek and why military service specifically fits your considered choice. Acknowledge that civilian R&D is also valuable service.

Supporting knowledge: Do not claim that patriotism belongs only to the armed forces.

Cross-questions and support: Why not continue engineering? What are you giving up? When did this become a serious goal?

Sources: [PIQ], [CERT], [SEL].

P024 | Why an officer’s role?

Your answer preparation: Discuss responsibility for people, decisions, training, resources and accountability using your own examples. Avoid status, uniform or authority as the sole reason.

Supporting knowledge: An engineering job title does not establish military command experience.

Cross-questions and support: What does responsibility mean when an outcome is poor? Would you accept routine administration?

Sources: [PIQ], [CERT], [SEL].

P025 | Why SSC non-technical despite your technical background?

Your answer preparation: Use the actual entry you applied for, your understanding of it and your career preference. Know other eligible routes without inventing eligibility.

Supporting knowledge: Your PIQ says UPSC and SSC(NT) Men-125. The applicable notification controls eligibility and terms.

Cross-questions and support: Did you consider technical entries? Is branch allocation guaranteed? What if assigned a different role?

Sources: [PIQ], [CERT], [SEL].

P026 | What do you know about your course and academy?

Your answer preparation: Read your exact notification and call-up letter. Prepare course title, academy, training, commission terms and reporting instructions from those records.

Supporting knowledge: Do not use NCC-125 material simply because the number 125 matches.

Cross-questions and support: What documents are required? What is the difference between recommendation and final joining?

Sources: [PIQ], [CERT], [SEL].

P027 | What if you are not recommended?

Your answer preparation: State a practical plan: continue work, review preparation, maintain fitness and apply only to entries for which you are eligible.

Supporting knowledge: No previous result is established; give actual attempt history if asked.

Cross-questions and support: What would you change? Would you resign to prepare? How would you know what needs work?

Sources: [PIQ], [CERT], [SEL].

P028 | Which arm or service interests you, and why?

Your answer preparation: Study the public role of your preference, then explain genuine reasons. Keep willingness to serve beyond a preferred posting.

Supporting knowledge: Do not assume mountaineering or software experience guarantees a particular arm.

Cross-questions and support: What if you do not get it? How is your background relevant without being decisive?

Sources: [PIQ], [CERT], [SEL].

P029 | How have you prepared for separation from family and field conditions?

Your answer preparation: Discuss actual time away, courses, independent living and financial planning. Acknowledge that civilian adventure is not the same as military duty.

Supporting knowledge: BMC dates and grade are documented; your adaptation narrative is yours.

Cross-questions and support: What did you find difficult during NIMAS? What support arrangements exist at home?

Sources: [PIQ], [CERT], [SEL].

P030 | Why did you not join NCC, if you did not?

Your answer preparation: First confirm the actual PIQ entry. Give the real availability, timing or decision, without presenting NSS as military training.

Supporting knowledge: The visible NCC field is not reliably resolved by extraction; do not assume absence.

Cross-questions and support: What did you choose instead? What is NSS versus NCC?

Sources: [PIQ], [CERT], [SEL].

P031 | What does discipline mean when nobody is checking?

Your answer preparation: Use a small repeated example: training log, test documentation, safety checks or a financial commitment.

Supporting knowledge: Avoid claiming perfection or rehearsed heroic behaviour.

Cross-questions and support: When have you broken your own routine? How did you recover?

Sources: [PIQ], [CERT], [SEL].

P032 | How will you leave your current team responsibly if selected?

Your answer preparation: Prepare notice, handover, documentation and confidentiality considerations under your actual contract.

Supporting knowledge: Discuss process generically; do not disclose proprietary project details.

Cross-questions and support: Would you share code at interview? What if a deadline overlaps reporting?

Sources: [PIQ], [CERT], [SEL].

6 | Category P: current job, internships and research ownership

P033 | Explain your current job to a farmer in one minute.

Your answer preparation: Use your actual scope: technical research, mathematical analysis and path-planning algorithms for autonomous tractors. Explain the practical goal before naming algorithms.

Supporting knowledge: Your message establishes the current role; certificate 08 establishes the earlier ROS simulation training only.

Cross-questions and support: What does the tractor receive from your module? Does it control steering? What remains outside your responsibility?

Sources: [PIQ], [CERT].

P034 | Walk me through one algorithm you personally developed.

Your answer preparation: Use a non-confidential example: problem, inputs, constraints, method, outputs, tests and limitations. Separate your work from team integration.

Supporting knowledge: Use the technical cards for general principles; do not present illustrative numbers as company results.

Cross-questions and support: Why this method? What alternative did you reject? What happens when no feasible solution exists?

Sources: [PIQ], [CERT].

P035 | What was your most difficult path-planning case?

Your answer preparation: Choose a real case you may discuss: irregular boundary, narrow headland, obstacle, implement off-tracking or localisation degradation. Explain diagnosis and correction.

Supporting knowledge: The attachments do not identify a specific production failure or improvement percentage.

Cross-questions and support: How did you reproduce it? What regression tests did you add? Did the fix affect other cases?

Sources: [PIQ], [CERT].

P036 | How do you know an algorithm is better?

Your answer preparation: State the actual baseline, metrics, dataset scope and validation conditions. Discuss failures as well as averages.

Supporting knowledge: A shorter path alone is not proof of safer or better coverage.

Cross-questions and support: Was the comparison fair? What is worst-case performance? Was it tested outside the simulator?

Sources: [PIQ], [CERT].

P037 | What do you research before writing code?

Your answer preparation: Describe actual literature search, problem formulation, assumptions, feasibility checks and prototype process.

Supporting knowledge: Keep a list of public references you genuinely read; do not claim papers from this handbook as prior reading.

Cross-questions and support: How do you judge a paper? What if results do not reproduce?

Sources: [PIQ], [CERT].

P038 | What did you do in the ROS tractor-simulation internship?

Your answer preparation: Prepare simulator platform/version, vehicle model, interfaces, controller, test cases and your own contribution.

Supporting knowledge: Certificate 08 says design/development of simulation for a self-guided tractor with control system using ROS, April-August 2025; versions are not specified.

Cross-questions and support: Was it ROS 1 or ROS 2? Which simulator? Was hardware involved?

Sources: [PIQ], [CERT].

P039 | What did you build at the Virtual Labs project?

Your answer preparation: Prepare actual experiment names, user interactions, mathematical model, validation, browser testing and accessibility considerations.

Supporting knowledge: Certificate 01 lists Engineering Graphics Virtual Lab work under Virtual Labs III (Extended).

Cross-questions and support: What makes a virtual experiment educational? How did you check numerical correctness?

Sources: [PIQ], [CERT].

P040 | How did you balance Virtual Labs, IARI and academic work?

Your answer preparation: Prepare an honest calendar with hours and on-site/remote arrangements. Explain documented gaps without filling them by assumption.

Supporting knowledge: The certificate lists discrete engagements, not uninterrupted full-time employment.

Cross-questions and support: Did engagements overlap? Who approved the arrangement? Why did remuneration change?

Sources: [PIQ], [CERT].

P041 | What did you learn at IARI that helps your current role?

Your answer preparation: Choose a real component, CAD/manufacturing exercise or mechatronics task. Explain design requirements and validation.

Supporting knowledge: Certificate 09 supports June 2024 training in additive manufacturing and mechatronics for farm machinery design.

Cross-questions and support: Which material and process? What tolerance mattered? How would you manufacture it at scale?

Sources: [PIQ], [CERT].

P042 | What is the difference between your job and your internship?

Your answer preparation: Compare actual ownership, review, deliverables and employment terms without assessing colleagues.

Supporting knowledge: PIQ lists current R&D Engineer and previous project intern; joining date is not supplied.

Cross-questions and support: What can you now do independently? What still requires review?

Sources: [PIQ], [CERT].

P043 | Describe a technical disagreement without criticising a colleague.

Your answer preparation: Explain the design issue, evidence you collected, decision process and your action after the decision.

Supporting knowledge: Do not speculate about anyone’s motives, competence or attitude.

Cross-questions and support: What if your proposal was not chosen? How did you test both options?

Sources: [PIQ], [CERT].

P044 | What would you refuse to disclose in this interview?

Your answer preparation: Identify categories: source code, unreleased specifications, confidential test data and customer information. Offer a generic public-domain explanation instead.

Supporting knowledge: Prepare a sanitised example that is genuinely independent of proprietary material.

Cross-questions and support: Can you explain the mathematics without company data? What can you draw safely?

Sources: [PIQ], [CERT].

7 | Category P: leadership, service and extracurricular depth

P045 | What did you actually do as training and placement coordinator?

Your answer preparation: Prepare tenure, routine tasks, stakeholders, a scheduling issue, your action and documented outcome.

Supporting knowledge: Certificate 02 supports coordination, communication and programme support, not authority over recruitment decisions.

Cross-questions and support: How many events did you support? Did you select candidates? How did you protect student information?

Sources: [PIQ], [CERT].

P046 | What did class captain involve?

Your answer preparation: State exact class/year, selection process, tasks and one problem resolved.

Supporting knowledge: The PIQ responsibility field is truncated; confirm title and tenure.

Cross-questions and support: Were you elected? What happened when a class request was unreasonable?

Sources: [PIQ], [CERT].

P047 | Describe your NSS special camp.

Your answer preparation: Recall location, daily schedule, community needs, your tasks and follow-up. Separate organising from clinical work.

Supporting knowledge: Certificate 03: 240 hours over 2022-24 and camp 18-24 December 2022.

Cross-questions and support: What did you do on a particular day? How did you measure usefulness?

Sources: [PIQ], [CERT].

P048 | What was your role in weekly medical camps?

Your answer preparation: Describe registration, queue management, logistics or awareness work actually performed. Do not imply diagnosis or treatment without qualification.

Supporting knowledge: PIQ mentions weekly medical camps; it does not establish medical training.

Cross-questions and support: How did you handle privacy? What if someone asked you for medicine?

Sources: [PIQ], [CERT].

P049 | Why did you donate blood?

Your answer preparation: Give your real motivation and experience of the organised camp. Keep medical decisions with qualified staff.

Supporting knowledge: Certificate 16 is dated 15 October 2025; donor medical details are not needed here.

Cross-questions and support: Who organised it? Would you volunteer even if unable to donate?

Sources: [PIQ], [CERT].

P050 | What did your hackathon team attempt in 2023?

Your answer preparation: Prepare team SRAJAN, problem, proposed solution, your role, prototype and actual selection stage.

Supporting knowledge: Certificate 14 is internal participation, not evidence of national selection.

Cross-questions and support: What worked? What was not built? What feedback changed the design?

Sources: [PIQ], [CERT].

P051 | What changed in the 2024 Agri_Catalysts attempt?

Your answer preparation: Explain whether it was a new problem or iteration. Identify your own work and evidence.

Supporting knowledge: Certificate 13: internal hackathon on 10 September 2024.

Cross-questions and support: Why a different team? What was the outcome? Did users test it?

Sources: [PIQ], [CERT].

P052 | What assistance did you provide at the fNIRS workshop?

Your answer preparation: List actual tasks and what you learned at an introductory level. Avoid claiming you conducted a brain study.

Supporting knowledge: Certificate 15 records assistance, 3-7 March 2025.

Cross-questions and support: What is fNIRS? What does it measure indirectly? What was your technical versus logistical role?

Sources: [PIQ], [CERT].

P053 | Which DIY project can you explain from parts to testing?

Your answer preparation: Choose a real project. Prepare objective, parts, cost, circuit/mechanics, software, safety and one failed iteration.

Supporting knowledge: No DIY project title is provided; do not invent a robotics build.

Cross-questions and support: Why build instead of buy? What would you change? What did you measure?

Sources: [PIQ], [CERT].

P054 | What did your AI and IIRS courses change in your work?

Your answer preparation: Name one concept applied and one you have not applied. Separate course completion from practical expertise.

Supporting knowledge: AI module: 85% deemed percentage. IIRS merit course: grade B and 73% attendance, not 73% marks.

Cross-questions and support: What was graded? What was attendance only? How would you demonstrate learning?

Sources: [PIQ], [CERT].

8 | Category P: mountaineering, running, sport and travel

P055 | Why did you take the NIMAS basic course?

Your answer preparation: Explain your actual motivation, preparation, funding and choice of institute.

Supporting knowledge: Certificate 04: 23 May-19 June 2026, A grade, advanced-course recommendation.

Cross-questions and support: Why Dirang? What did you expect versus experience? What will you do next?

Sources: [PIQ], [CERT].

P056 | Describe the hardest day on your course.

Your answer preparation: Use actual route, weather, task and your response. Avoid inventing rescue stories or summit achievements.

Supporting knowledge: Highest altitude actually reached and route are not established by the certificate.

Cross-questions and support: What equipment did you carry? What did you learn? When would you turn back?

Sources: [PIQ], [CERT].

P057 | What does your A grade establish?

Your answer preparation: Explain the course result and recommendation exactly as printed. Distinguish assessment from a permanent safety guarantee.

Supporting knowledge: The certificate’s expedition suitability wording is not proof of a 6,500 m ascent.

Cross-questions and support: Have you completed advanced training? Have you climbed a 6,500 m peak?

Sources: [PIQ], [CERT].

P058 | Why sport climbing after mountaineering?

Your answer preparation: Describe actual skill goals and what Level 1 added. Prepare venue, instructor-led activities and techniques practised.

Supporting knowledge: Certificate 05 supports IMF Level 1 with A grade; confirm handwritten date.

Cross-questions and support: What is lead versus top-rope? What can you not safely teach independently?

Sources: [PIQ], [CERT].

P059 | Compare your 10 km and half-marathon experiences.

Your answer preparation: Use actual pacing, preparation, conditions and recovery. Explain why pace differed without claiming an unrecorded cause.

Supporting knowledge: 10 km: 53:32 on 22 February 2026. Printed 21 km: 2:17:01 in the Faridabad event.

Cross-questions and support: Did you start too fast? What is your next target? How do you balance running with strength work?

Sources: [PIQ], [CERT].

P060 | What does position 287 mean?

Your answer preparation: State only the printed position and say whether you know the category and field size.

Supporting knowledge: The certificate does not supply a denominator or rank category, so no percentile can be calculated.

Cross-questions and support: How many participants? What percentile? Was it age-category or overall?

Sources: [PIQ], [CERT].

P061 | What is your role in volleyball?

Your answer preparation: Prepare actual position, frequency, department event and one tactical decision you made.

Supporting knowledge: PIQ suggests department/company representation; it does not establish district or state level.

Cross-questions and support: How does rotation work? What do you do when receiving serve?

Sources: [PIQ], [CERT].

P062 | Why did you stop or reduce cricket after school?

Your answer preparation: Use the actual reason and current participation, if any.

Supporting knowledge: PIQ shows school participation around 2018-19; confirm dates.

Cross-questions and support: What position did you play? What was your best documented contribution?

Sources: [PIQ], [CERT].

P063 | What did travel to Arunachal and Amritsar teach you?

Your answer preparation: Discuss actual observations, route and practical adaptations, not stereotypes about communities.

Supporting knowledge: Course and race locations support these question routes, not a complete travel itinerary.

Cross-questions and support: What places did you visit? What did you verify afterward?

Sources: [PIQ], [CERT].

P064 | How would you help a teammate while protecting group safety?

Your answer preparation: Choose an actual training example if available. Explain communication, instructor involvement and limits of your competence.

Supporting knowledge: Use instructor-led safety principles; a certificate is not authority to conduct independent rescue.

Cross-questions and support: What if the person wants to continue despite risk? Would you abandon a summit?

Sources: [PIQ], [CERT].

9 | Category F: Sarendhi, Agra, Mathura and regional awareness

F001 | Give a concise historical account of Agra.

Factual answer / explanation: Agra is a Yamuna-side city whose prominence grew under the Lodi and Mughal rulers. Sikandar Lodi established his capital there in 1504; Mughal rule made it a major imperial centre. Akbar, Jahangir and Shah Jahan are central to its architectural history. British control began in 1803. In interview, separate the history of the settlement from the later creation of a capital or cantonment.

Supporting knowledge: For Sarendhi, prepare your actual gram panchayat, block, tehsil, nearest town, road connections, local crops, irrigation sources and one civic problem. These village-level details were not reliably verified, so no local officeholder or population is invented.

Cross-questions and support: Who preceded the Mughals? The Lodi dynasty. Why does a river matter? As an analytical explanation, water supply, agriculture, transport and defensible crossings can support urban growth. Is all Agra a cantonment? No: the notified cantonment is a specific area.

Sources: [AGH], [CANT], [PIQ].

F002 | Which UNESCO sites are associated with Agra?

Factual answer / explanation: Taj Mahal and Agra Fort were inscribed in 1983; Fatehpur Sikri in 1986. The Taj Mahal is a white-marble mausoleum commissioned by Shah Jahan. Agra Fort is a major Mughal fortified complex, not the Red Fort in Delhi. Fatehpur Sikri was built under Akbar and served as an imperial capital.

Supporting knowledge: Avoid treating every popular construction anecdote as established history.

Cross-questions and support: Who built which? Shah Jahan is associated with the Taj; Akbar with the development of Fatehpur Sikri. Are all three in Agra city centre? No: distinguish the city from the wider district/region. Why different Taj completion dates? Sources may refer to the mausoleum versus the wider complex; UNESCO describes 1631-1648 for the mausoleum.

Sources: [HER], [TAJ], [FS].

F003 | Why is Agra important beyond the Taj Mahal?

Factual answer / explanation: Its economy includes tourism, footwear/leather products, handicrafts and foundry-related activity. Its defence relevance includes Agra Cantonment, an air-force station, parachute training and ADRDE. This gives you a stronger answer than listing monuments alone.

Supporting knowledge: A local-development answer can connect your engineering work to water, waste, transport or small-enterprise productivity. Do not invent current pollution readings or employment shares.

Cross-questions and support: How would you improve the city? Personal judgement: identify one issue, evidence, feasible intervention and trade-off. Can industry and heritage coexist? Discuss pollution control, cleaner processes and enforcement without claiming a particular scheme has succeeded.

Sources: [AGE], [CANT], [ADR], [PTS].

F004 | Is Agra itself a cantonment?

Factual answer / explanation: Agra contains a notified cantonment; the whole city is not synonymous with that cantonment. The Agra Cantonment Board states that it was established in 1805 and administers municipal, revenue, land and development matters in its area, serving civilian residents and Army units.

Supporting knowledge: Do not memorise changing ward counts or civil-area transfer status without checking the latest official notification.

Cross-questions and support: Cantonment versus military station? Explain that a notified cantonment has a statutory civic-administration framework; a military station is a military establishment and need not be identical to a cantonment. Who manages the whole city? Do not confuse the cantonment board with the municipal corporation.

Sources: [CANT].

F005 | What is ADRDE and why is it in your hometown preparation?

Factual answer / explanation: ADRDE stands for Aerial Delivery Research and Development Establishment, a DRDO establishment in Agra. Its public mandate includes parachute and aerial-delivery systems, aircraft arrester barriers and lighter-than-air systems. It is a research and development establishment, not a parachute training school.

Supporting knowledge: Do not discuss unpublished specifications, locations inside facilities or operational deployment details.

Cross-questions and support: DRDO versus a manufacturer? R&D develops and validates technology; production can involve industry partners. ADRDE versus PTS? One develops systems, the other trains personnel. How is your work related? Only at the general level of modelling, control, navigation and validation, not identical missions.

Sources: [ADR].

F006 | What is the Paratroopers Training School at Agra?

Factual answer / explanation: The PTS is an Indian Air Force training establishment associated with parachute training, including training Army personnel. The specialist IAF unit database places it at Agra and gives a raising date of 14 August 1947. Some secondary accounts use 1948 for the Agra establishment; avoid presenting those dates as interchangeable.

Supporting knowledge: Know the distinction between airborne training, equipment R&D and an operational airborne formation.

Cross-questions and support: Is it an Army or Air Force school? The organisational affiliation is IAF even though Army personnel train there. Is a paratrooper automatically special forces? No: parachute qualification and special-forces selection are different concepts. Is a parachute jump the same as mountaineering? No; skills and training requirements differ.

Sources: [PTS].

F007 | Which corps headquarters is at Mathura?

Factual answer / explanation: I Corps, commonly called Strike One, is headquartered at Mathura. It was raised on 1 April 1965 at Varanasi. PTI’s historical account supports the raising date and Mathura association. A 2 October 2026 report identifies Lt Gen Akhilesh Kumar as the new GOC and places the corps under Northern Command. Treat the current command detail as dated secondary reporting.

Supporting knowledge: Avoid reproducing current order of battle, exact strength, readiness or deployment details. The public headquarters and broad history are enough for this interview route.

Cross-questions and support: Was it raised at Mathura? No: the cited historical account says Varanasi. Why might older notes say South Western Command? Organisational assignments change; the 2026 report describes the 2021 move to Northern Command. What is a corps? A large Army formation; do not give a universal fixed troop count.

Sources: [CORPS], [CORPS26].

F008 | How would you connect Agra, Mathura and your engineering background?

Factual answer / explanation: A factual base is Agra's aerial-delivery R&D and parachute training, Mathura's corps headquarters and your documented agricultural-engineering education. The connection is an analytical one: different organisations require reliable systems, logistics, communication, safety and disciplined testing. Your experience does not establish military operational expertise.

Supporting knowledge: Prepare a simple map from memory showing Agra, Mathura, Faridabad and Prayagraj, with approximate directions. Use an authoritative map to verify distances rather than guessing.

Cross-questions and support: Can tractor autonomy be transferred directly to military use? No: requirements, terrain, communications, threats, certification and human authority differ. What is transferable? General modelling, software verification, navigation and failure-analysis skills.

Sources: [ADR], [PTS], [CORPS], [PIQ].

F009 | What should you know about Faridabad?

Factual answer / explanation: Faridabad is in Haryana and is strongly associated with industry. The district court history dates the city's founding to 1607 under Shaikh Farid and the creation of the separate district to 15 August 1979. The district portal currently displays 1707 for the founding while also associating it with Jahangir, so there is a source discrepancy. Prefer the court history for the 1607 account and acknowledge the inconsistency if pressed.

Supporting knowledge: Prepare actual commute, industrial locality, civic issue and local transport you use. Do not claim a residence based solely on employer location.

Cross-questions and support: Is 1,809,733 a current city population? No: distinguish district statistics from city population and date them. Why is it relevant to you? EKL training was at Faridabad, but current residence still requires your confirmation.

Sources: [FAR], [FAR2], [CERT].

F010 | What should you know about Prayagraj, your selection-centre city?

Factual answer / explanation: Prayagraj lies at the confluence of the Ganga and Yamuna; the traditional Triveni also refers to the Saraswati. The district history discusses the Mughal fort, British-era history and its role in the freedom movement. The Army overview identifies Selection Centre East at Prayagraj, while your PIQ uses Allahabad.

Supporting knowledge: Do not confuse the selection centre with every defence establishment in the city.

Cross-questions and support: Are Allahabad and Prayagraj different cities? They refer to the same city under different names. What is your reporting point? Use your current call-up letter, not a generic old instruction sheet. What should you visit? Only if permitted and consistent with reporting and testing requirements.

Sources: [PRAY], [SEL], [PIQ].

F011 | How should you answer a population or local-administration question?

Factual answer / explanation: Give the geographic unit, reference year and source before the number. A village, municipal city, urban agglomeration and district are different units. Your PIQ's approximate Sarendhi population is 10,000 and present-residence population 1,590,000; these are your declared approximations, not independently verified current census figures.

Supporting knowledge: Personal worksheet: village/block/tehsil; district/state; constituency; local water source; two crops; nearest hospital; route to selection centre. No officeholder names are fabricated here.

Cross-questions and support: Who is your MP, MLA, district magistrate or panchayat head? Confirm the correct constituency and current official listing before interview. Which level handles drainage or roads? First identify whether the issue is within panchayat, municipality, cantonment or another authority.

Sources: [PIQ], [FAR2], [CANT].

10 | Category F: institutions, employer and professional awareness

F012 | What is distinctive about Dayalbagh Educational Institute?

Factual answer / explanation: DEI is in Dayalbagh, Agra. Its institutional history describes an integrated educational approach and deemed-university status in 1981. Keep the institute's legal status, the earlier educational movement and your own faculty/programme distinct.

Supporting knowledge: Your certificate 02 records B.Tech Agricultural Engineering, 2022-2026. Your PIQ gives English as medium and day-scholar status.

Cross-questions and support: Does deemed university mean a school board? No: do not confuse university-level status with UP Board school examinations. What did you experience personally? Supply your actual coursework, practical work and service activities.

Sources: [DEI], [DEI2], [PIQ], [CERT].

F013 | What is ICAR-IARI and what did its engineering division do?

Factual answer / explanation: IARI is the Indian Agricultural Research Institute. Its Agricultural Engineering Division states that it was established in 1945. Its mandate covers farm machinery and power, processing and structures, protected cultivation, soil/water engineering and renewable energy, with research, teaching, testing and extension functions.

Supporting knowledge: Your actual project details, material, machine and measurements must come from your internship records.

Cross-questions and support: ICAR versus IARI? ICAR is the council; IARI is an institute. Was your internship an agricultural degree? No: your certificate establishes a June 2024 internship. How does machinery testing differ from design? Testing evaluates performance against requirements; design creates the solution.

Sources: [IARI], [CERT].

F014 | Give an accurate public overview of Escorts Kubota.

Factual answer / explanation: The FY2025-26 integrated annual report describes two core business segments: Agri Machinery and Construction Equipment. Farmtrac, Powertrac and Kubota form its three-brand tractor strategy. It records the 2022 name change to Escorts Kubota Limited and the 2024 amalgamation of two Kubota joint ventures.

Supporting knowledge: The report names Nikhil Nanda as Chairman & Managing Director and Akira Kato as Deputy Managing Director. These are public appointments, not claims about your reporting line.

Cross-questions and support: Does it still have a continuing railway-equipment division? The report says the railway business transfer was completed on 1 June 2025 and is treated as discontinued operations. Why does this matter? Old company profiles can be out of date.

Sources: [EKL].

F015 | What company figures can you quote responsibly?

Factual answer / explanation: For FY2025-26, the annual report gives total tractor volume of 133,670 units and standalone revenue from continuing operations of Rs 11,472.8 crore. It reports continuing-operations PAT of Rs 1,381.0 crore and reported PAT including discontinued operations of Rs 2,408.6 crore. Do not mix the two profit bases.

Supporting knowledge: The report also gives approximately 10.9% market share; if quoting it, retain the report's period and context rather than calling it today's share.

Cross-questions and support: Revenue versus profit? Revenue is sales/income from operations; profit is after relevant expenses. Why can reported profit differ from continuing profit? Discontinued operations can affect the reported total. Are these your project's results? No: company-level figures cannot establish your individual impact.

Sources: [EKL].

F016 | What is RNIL and what can you say about autonomous tractors publicly?

Factual answer / explanation: The annual report identifies RNIL as the Rajan Nanda Innovation Lab, launched in 2020 to nurture new and disruptive business ideas. Its history records an autonomous tractor concept in 2018. A concept milestone is not proof that every present autonomous capability is commercially deployed.

Supporting knowledge: Your certificate 08 establishes training in the RNIL R&D division and a ROS simulation project. Your current role is supplied by you, not independently certified in that document.

Cross-questions and support: Can you state your product's release date or accuracy? Only if publicly released and authorised. How do you discuss your role? Describe general research, mathematical analysis and planning, then clearly mark your actual non-confidential contribution.

Sources: [EKL], [CERT].

F017 | How do farm mechanisation and sustainable mechanisation differ?

Factual answer / explanation: FAO describes mechanisation broadly, from hand tools to sophisticated motorised equipment. Sustainable mechanisation also considers economic, social, environmental and cultural factors. It is not simply maximising machine size or replacing all labour.

Supporting knowledge: For your interview, connect technology to timeliness, resource efficiency, operator burden and maintenance, not just automation novelty.

Cross-questions and support: Why might a small farmer not buy an autonomous tractor? Suggested analysis: purchase cost, utilisation, field fragmentation, service access and trust. What alternative could work? Shared or hire-service models may spread cost, but viability requires local evidence.

Sources: [FAO].

F018 | What is the difference between a prototype, simulation and validated product?

Factual answer / explanation: A simulation evaluates a model in a virtual environment. A prototype is an early implementation used to test ideas. Validation asks whether a system meets the intended use and requirements in relevant conditions. A successful simulated run does not establish safe real-world deployment.

Supporting knowledge: This distinction is particularly relevant because your EKL certificate explicitly describes simulation.

Cross-questions and support: What is verification? Checking that the implementation meets its specification. What is the simulation-to-reality gap? Differences between model assumptions and physical behaviour. How would you reduce it? Suggested engineering approach: calibrated models, controlled tests and traceable acceptance criteria.

Sources: [CERT], [CONTROL], [ROBOT].

11 | Category F: agricultural and mechanical engineering fundamentals

F019 | What does an agricultural engineer contribute?

Factual answer / explanation: The field applies engineering to production and post-production agriculture: machinery and power, irrigation/drainage, processing, storage, structures, energy and automation. IARI and ICAR describe these as connected technical areas. An agricultural scientist and an agricultural engineer can work on the same farming problem with different specialist methods.

Cross-questions and support: Give an example? For sowing, agronomy informs crop requirements while engineering addresses metering, placement, traction, power and machine control. Is bigger machinery always better? No: match it to field, task, power, cost and operating conditions.

Sources: [IARI], [ICAR], [FAO].

F020 | Define theoretical field capacity, effective field capacity and field efficiency.

Factual answer / explanation: For width W in metres and speed v in km/h, theoretical field capacity TFC = W*v/10 hectares/hour. Effective field capacity EFC = actual completed area / field time. Field efficiency = EFC/TFC, expressed as a fraction or percentage. Turning, overlap and in-field delays reduce effective capacity.

Supporting knowledge: All example numbers are hypothetical teaching values, not EKL test results.

Cross-questions and support: Illustrative calculation? W=2.4 m and v=5 km/h gives TFC=1.2 ha/h. At 75% field efficiency, EFC=0.9 ha/h. How long for 4.5 ha? 4.5/0.9=5 hours under those assumptions. Are travel-to-field delays included? The cited teaching note distinguishes outside-field delays from field efficiency.

Sources: [FARM].

F021 | What is draft and how does it relate to power?

Factual answer / explanation: Draft is the horizontal component of implement pull in the direction of travel. Mechanical power associated with pull is force times speed: P=F*v. In SI units, newtons times metres/second gives watts. A tractor must also meet traction, stability and implement-interface requirements, not just engine-power demand.

Cross-questions and support: Illustrative calculation? A 5,000 N draft at 1.5 m/s requires 7.5 kW at the drawbar, before considering other power demands and losses. Why can a powerful tractor still struggle? Poor traction, unsuitable gearing, soil conditions or implement mismatch.

Sources: [FARM], [ICAR].

F022 | What is the difference between engine, PTO and drawbar power?

Factual answer / explanation: Engine power is produced by the engine; PTO power is available through the power take-off for driven implements; drawbar power is associated with useful pulling force and travel speed. Losses and operating conditions mean these are not interchangeable ratings.

Supporting knowledge: Study the public specification of one tractor you genuinely know, rather than guessing your test vehicle’s unpublished rating.

Cross-questions and support: Which matters for a rotary implement? PTO demand is important, along with draft, lifting capacity and stability. Which matters for a trailed tillage tool? Pulling requirement and traction are central. Does the same horsepower guarantee identical field performance? No: matching and

conditions matter.

Sources: [ICAR], [FARM].

F023 | Explain torque, rotational speed and power.

Factual answer / explanation: For rotational motion, P=T*omega, where torque T is in N m and angular speed omega is in rad/s. If speed is N rpm, omega=2*pi*N/60. High torque and high power are related but not identical: power also depends on rotational speed.

Supporting knowledge: This is a standard mechanics derivation, not a tractor-specific specification.

Cross-questions and support: Illustrative calculation? At 100 N m and 1,500 rpm, power is approximately 15.71 kW. Why use gears? They trade output speed and torque subject to losses; they do not create energy. What is a unit check? N m multiplied by 1/s gives watts.

Sources: [ICAR].

F024 | Why does wheel slip matter?

Factual answer / explanation: Wheel slip reflects the difference between ideal rolling motion and actual ground progress. It affects traction, odometry and the relationship between wheel rotation and travelled distance. A planner/controller that assumes perfect rolling can accumulate error on soil.

Supporting knowledge: Suggested validation: test on different soil conditions and loads, with a consistent slip definition.

Cross-questions and support: How would you measure it? Compare a clearly defined theoretical wheel-based speed with an independent ground-speed estimate. Why state the convention? Different formula definitions can use different denominators. Does GNSS remove all slip problems? It helps observe motion but does not eliminate traction limits.

Sources: [ROBOT], [CONTROL], [ICAR].

F025 | What is mechatronics in a tractor?

Factual answer / explanation: Mechatronics integrates mechanical systems, electronics, sensing, control and software. A steering-control example combines a mechanical steering mechanism, actuator, sensors, controller and feedback. Your IARI certificate connects mechatronics to farm-machinery design.

Cross-questions and support: Sensor versus actuator? A sensor measures; an actuator produces a physical action. Open-loop versus closed-loop? Closed-loop uses feedback to correct error. Can software compensate for any mechanical defect? No: physical limits and faults still constrain operation.

Sources: [CERT], [ICAR], [CONTROL].

F026 | What is additive manufacturing, and what are its limitations?

Factual answer / explanation: Additive manufacturing builds a part by adding material, commonly layer by layer, rather than primarily removing it. In your internship discussion, explain the actual process used rather than assuming every 3D printer works the same way. Suggested design considerations include material, orientation, tolerance, surface finish, strength and repeatability.

Supporting knowledge: The certificate establishes the training theme, not the process, material or part you used.

Cross-questions and support: Why use it for a prototype? Suggested answer: rapid iteration and complex geometry can be useful. Why not use every printed part on a tractor? Functional loads, temperature, durability and qualification must be demonstrated. Did you use FDM, resin or metal printing? Only your records can answer.

Sources: [CERT], [ICAR].

F027 | How do you evaluate an autonomous tractor’s value to a farmer?

Factual answer / explanation: Use a complete comparison: task quality, effective field capacity, fuel/input use, supervision, downtime, service cost and total cost over the useful period. FAO’s sustainable-mechanisation framework supports considering economic and social factors as well as technical output.

Supporting knowledge: This is a suggested evaluation framework, not a claim that your company has achieved a particular saving.

Cross-questions and support: Is labour saving enough? Not if capital and maintenance costs outweigh benefits. What is a fair baseline? The same task, field conditions, implement and quality requirement. What evidence is needed? Measured operating data and transparent assumptions.

Sources: [FAO], [FARM].

12 | Category F: autonomy architecture and path planning

F028 | Distinguish perception, localisation, planning and control.

Factual answer / explanation: Perception interprets the environment; localisation estimates where the vehicle is; planning chooses a feasible route or motion; control commands actuators to follow the intended motion. Feedback control needs an estimate of vehicle configuration. These functions interact but should not be treated as one algorithm.

Supporting knowledge: Suggested generic flow: environment and mission inputs -> state estimation and constraints -> plan -> tracking controller -> actuator feedback. This is an illustrative architecture, not your employer’s design.

Cross-questions and support: Does a path planner steer the tractor? Not necessarily: it may output a path to a separate controller. Does localisation detect obstacles? Not by itself. What is your module boundary? Supply your actual interface and responsibility.

Sources: [CONTROL], [ROBOT].

F029 | What is a path, a trajectory and a coverage plan?

Factual answer / explanation: A path describes geometric motion; a trajectory adds timing or speed along that motion. Coverage planning seeks to work an area, unlike a single start-to-goal route. For a tractor, a usable plan must consider heading, turning limits, implement footprint and obstacles, not merely a line through points.

Supporting knowledge: The coverage-specific explanation is an engineering application of vehicle constraints, not a description of a certified EKL system.

Cross-questions and support: Why is shortest path insufficient? It may miss area or be infeasible for the vehicle. Can a geometrically smooth path still fail? Yes, if speed, steering rate, clearance or implement motion is not feasible.

Sources: [ROBOT], [CONTROL], [ASTAR].

F030 | What is a nonholonomic constraint?

Factual answer / explanation: A conventional wheel rolling without sideways slip cannot move instantaneously in every direction. A car-like vehicle must manoeuvre forward/backward and turn rather than translate sideways at will. Modern Robotics models this using configuration-dependent allowable velocities.

Cross-questions and support: Can a tractor shift sideways like a point on a screen? Not under the simple no-side-slip model. Does nonholonomic mean it can never reach a nearby sideways position? No: a sequence of manoeuvres can reach it. Why include heading in planning? Position alone does not capture feasible motion.

Sources: [ROBOT].

F031 | Explain the kinematic bicycle model.

Factual answer / explanation: A common simplified car-like model uses rear-axle reference position x,y, heading theta, wheelbase L, forward speed v and steering angle delta: x_dot=v*cos(theta), y_dot=v*sin(theta), theta_dot=v*tan(delta)/L. It assumes planar motion and a simplified rolling model.

Supporting knowledge: Illustrative example: L=2.5 m, delta=30 degrees gives R about 4.33 m. This is not a measured tractor specification.

Cross-questions and support: Turning radius? R=L/tan(delta) for the rear-axle reference in this ideal model. Why not use it blindly? Slip, actuator dynamics, articulation and terrain can invalidate assumptions. Is steering-wheel angle equal to tyre angle? Not generally; calibration and steering geometry matter.

Sources: [ROBOT].

F032 | What is a graph in path planning?

Factual answer / explanation: A graph has nodes and edges. Nodes represent states or locations; edges represent allowed transitions and can have costs. A map is not automatically a graph: you choose how to represent connectivity, movement and constraints.

Cross-questions and support: What should a node contain? Depending on the problem, position, heading or other state variables. Why do edge costs matter? They express what is being minimised, such as distance or time. Can a graph path violate vehicle limits? Yes, if the graph does not encode them.

Sources: [ASTAR].

F033 | Compare BFS, Dijkstra and A*.

Factual answer / explanation: BFS expands by steps and is suitable for equal-cost edges. Dijkstra expands by accumulated cost and handles non-negative varying costs. A* adds a heuristic estimate toward the goal, commonly f(n)=g(n)+h(n), where g is cost so far and h is estimated remaining cost.

Supporting knowledge: Do not claim A* directly solves complete agricultural coverage simply because it solves a start-to-goal search problem.

Cross-questions and support: When does A* resemble Dijkstra? When h=0. Is A* always faster? No: graph structure and heuristic quality matter. What does the heuristic need for an optimality guarantee? An admissible heuristic does not overestimate; implementation and consistency/reopening details also matter.

Sources: [ASTAR].

F034 | How do you choose a heuristic?

Factual answer / explanation: The heuristic should be a lower-bound estimate in the same cost units as g. Euclidean distance is a useful geometric lower bound when movement costs cannot be lower than straight-line distance. Manhattan distance fits certain axis-aligned grid assumptions but can overestimate when diagonal motion is allowed cheaply.

Supporting knowledge: The appropriate heuristic depends on the state space and cost model, not just the algorithm name.

Cross-questions and support: What if cost is time? Convert a distance lower bound using a valid maximum-speed bound, with consistent units. Can an aggressive heuristic sacrifice optimality? Yes. How do you check it? Compare against known small cases and a reference search.

Sources: [ASTAR].

F035 | Why can an ordinary grid path fail for a tractor?

Factual answer / explanation: A grid path may include abrupt heading changes or corners that a bounded-steering vehicle cannot follow. A finite vehicle and implement also occupy area, while a graph node may represent only a point. Vehicle-aware planning must respect feasible transitions and clearance.

Supporting knowledge: Suggested explanation: graph search proposes connectivity; motion feasibility and collision checking must still be represented or validated.

Cross-questions and support: Would smoothing solve everything? No: smoothing can move the path into an obstacle or outside the field. What must be rechecked? The full vehicle/implement footprint, turning limits and boundary constraints.

Sources: [ASTAR], [ROBOT].

F036 | How would you generate swaths in an irregular field?

Factual answer / explanation: Suggested engineering approach: define the workable region and exclusions, select a travel direction, construct spaced passes using effective implement width, clip them to the region, then connect feasible endpoints. Check coverage, overlap and turn feasibility together. This is a proposed reasoning framework, not a sourced company procedure.

Supporting knowledge: State assumptions about field geometry, implement offset and allowed overlap before discussing optimality.

Cross-questions and support: Why not choose the longest side automatically? It may not minimise turns, travel or soil impact. What about holes or concave boundaries? They can split passes and require separate connectivity decisions. What is effective width? The actual useful working width, which can differ from nominal width.

Sources: [FARM], [ROBOT].

F037 | What is a headland and why is width not one universal number?

Factual answer / explanation: In a coverage-planning discussion, the headland is the edge working/turning region. Required space depends on vehicle turning capability, implement dimensions, lane spacing, entry/exit headings and the boundary shape. A centreline fitting inside the field does not prove the complete machine fits.

Supporting knowledge: This is a general engineering explanation; do not present an empirical width formula as universally valid without its tested domain.

Cross-questions and support: Is twice the turning radius always enough? No universal guarantee follows, especially with implements and irregular boundaries. What should be checked? The swept footprint and allowed corridor. What if no feasible turn exists? Report infeasibility rather than silently crossing a boundary.

Sources: [ROBOT], [CONTROL].

F038 | How do you check a turn near a boundary?

Factual answer / explanation: Suggested method: represent the tractor and implement footprint along the full candidate motion, include the chosen clearance margin, and check containment in the allowed region and separation from obstacles. Checking only endpoints or sparse centreline samples can miss intermediate violations.

Supporting knowledge: A safe planning discussion should distinguish geometric feasibility, uncertainty margin and the independent stopping/supervision functions.

Cross-questions and support: What is swept area? The union of occupied footprints during motion. Why can a trailer cut inside? Its path can differ from the towing vehicle. How fine should sampling be? It must be justified against curvature, dimensions and required clearance, not chosen arbitrarily.

Sources: [ROBOT], [CONTROL].

F039 | How do coverage, overlap and missed area differ?

Factual answer / explanation: For a specified target region, covered area is the union of useful implement footprints intersected with that region. Missed area is the target minus that union. Overlap concerns repeated coverage; its definition must specify whether it counts multiply worked area once or counts extra passes multiplicatively.

Supporting knowledge: These are set-based mathematical definitions. Your actual software's metric must state its convention.

Cross-questions and support: Can adding pass areas overestimate coverage? Yes, overlapping footprints are double-counted. Can coverage exceed 100%? Unique target coverage cannot; a cumulative worked-area metric can. What about outside-field work? Report it separately, not as useful coverage.

Sources: [FARM].

F040 | How would you define an optimisation objective?

Factual answer / explanation: Suggested formulation: minimise a weighted combination of non-working distance, turns, time, overlap or other costs, subject to hard constraints for containment, clearance and vehicle feasibility. Keep safety-critical constraints hard rather than allowing a low weighted cost to justify a violation.

Supporting knowledge: Do not claim global optimality for a heuristic without proof or an appropriate bound.

Cross-questions and support: Why normalise terms? Metres, seconds and square metres have different scales. Is one optimum universal? No: it depends on objectives and assumptions. How do you show a trade-off? Compare feasible alternatives under the same conditions.

Sources: [ASTAR], [ROBOT].

F041 | How should a planner handle no feasible path?

Factual answer / explanation: Suggested design principle: return an explicit, diagnosable failure rather than an invalid path. Distinguish disconnected geometry, insufficient turning room, invalid inputs and a search timeout. A supervisor can then choose a safe response under the system's validated operating rules.

Supporting knowledge: This is a proposed software contract, not an assertion about EKL's implementation.

Cross-questions and support: Can a timeout prove no path exists? No: it may only mean the search budget was exhausted. Should it silently reduce safety margins? No. What should be logged? Input version, constraints, failure category and reproducible context, respecting confidentiality.

Sources: [ASTAR], [CONTROL].

F042 | How would you test path-planning software?

Factual answer / explanation: Suggested test matrix: rectangular, triangular, concave and holed fields; narrow passages; reversed polygon ordering; near-collinear edges; zero/invalid width; different turning radii; offset implements; and degraded localisation. Check invariants as well as visual appearance.

Supporting knowledge: Separate unit tests, integration tests, simulation and controlled physical validation. No test suite alone proves safety in every condition.

Cross-questions and support: What invariants? No forbidden boundary crossing, valid finite coordinates, continuity, feasible curvature and clearly defined coverage metrics. What is regression testing? Re-running earlier cases after a change. What is a fair benchmark? Identical inputs, constraints and metrics.

Sources: [ROBOT], [CONTROL], [ASTAR].

F043 | What is pure pursuit in simple terms?

Factual answer / explanation: As a conceptual tracking method, pure pursuit steers toward a look-ahead point on the desired path. In a common bicycle-model geometry, curvature is 2*sin(alpha)/Ld, where alpha is the target bearing relative to vehicle heading and Ld is look-ahead distance; steering follows delta=atan(L*curvature). This is a standard model-based explanation, not evidence that your project used it.

Supporting knowledge: State the coordinate and reference-point convention before using the formula.

Cross-questions and support: Short look-ahead? It can respond sharply but may be sensitive to noise. Long look-ahead? It can be smoother but cut corners. Why mention speed? Tracking behaviour and safe curvature/speed combinations are linked.

Sources: [ROBOT], [CONTROL].

F044 | What is the difference between planning and feedback control?

Factual answer / explanation: Planning provides intended motion; feedback control uses estimated current state and tracking error to correct commands. Modern Robotics notes that the planned motion should leave control authority for error correction rather than use all actuator capability continuously.

Supporting knowledge: Your role can be planning-focused while still requiring an understanding of control interfaces.

Cross-questions and support: Why can a perfect plan track poorly? State-estimation errors, disturbances, slip, delay or actuator limits. Can a controller fix an infeasible path? Not reliably: the planner must respect physical constraints.

Sources: [CONTROL].

13 | Category F: GNSS, sensors, ROS, AI and remote sensing

F045 | What is GNSS, and how is it different from GPS?

Factual answer / explanation: GNSS is the general term for satellite-navigation systems. GPS is the United States system; GLONASS is Russian, Galileo European and BeiDou Chinese. ISRO identifies NavIC as India’s regional system and QZSS as Japan’s regional system. Calling every satellite-navigation receiver simply GPS can hide multi-constellation capability.

Cross-questions and support: What does a receiver estimate? Position, velocity and time, depending on design and measurements. Why multiple constellations? They can improve satellite availability and geometry, but do not remove every error source.

Sources: [NAV2].

F046 | What is NavIC and why does it matter strategically?

Factual answer / explanation: NavIC means Navigation with Indian Constellation, formerly IRNSS. ISRO describes a regional service covering India and up to 1,500 km beyond its boundary, with Standard Positioning Service for civilians and Restricted Service for strategic users. Its nominal design uses seven satellites; that is a design description, not a verified count of healthy satellites today.

Supporting knowledge: Do not claim a current operational satellite count or guaranteed accuracy for your receiver without current system and equipment data.

Cross-questions and support: Does NavIC automatically give centimetre tractor guidance? No: standalone positioning and precision augmentation are different. Why sovereign navigation? ISRO notes the value of an independent system under Indian control. What bands? The service description includes L5 and S, with L1 introduced on newer satellites.

Sources: [NAV], [NAV2].

F047 | What is RTK and how is it different from ordinary standalone positioning?

Factual answer / explanation: Real-Time Kinematic positioning is a differential GNSS technique using carrier-phase measurements and reference information to improve precision. ESA distinguishes RTK, differential GNSS, SBAS and PPP as augmentation approaches. Performance depends on conditions, corrections, ambiguity resolution and equipment.

Supporting knowledge: No universal centimetre accuracy or correction-age threshold is claimed for your system.

Cross-questions and support: Fixed versus float? They refer to different ambiguity-resolution states; do not treat every fix as equally trustworthy. What if corrections stop? Accuracy and integrity can degrade; the vehicle’s validated supervisor must decide whether to continue, slow or stop.

Sources: [GNSS].

F048 | Why are accuracy and integrity different?

Factual answer / explanation: Accuracy concerns closeness to the true value. Integrity concerns confidence in the information and timely warning when it should not be used. An apparently smooth position can still be wrong. ESA’s augmentation overview explicitly distinguishes accuracy improvement from integrity support.

Cross-questions and support: Can low noise prove correctness? No: a biased signal can be precise but inaccurate. What matters near a boundary? The uncertainty and trustworthiness of the estimate, not just the displayed number of decimal places.

Sources: [GNSS].

F049 | What do IMU and wheel odometry contribute?

Factual answer / explanation: An IMU provides inertial measurements such as angular rate and acceleration; wheel odometry estimates motion from wheel movement. In a vehicle model, odometry can support state estimation, while external sensors can help correct accumulated error. Neither should be described as drift-free ground truth.

Supporting knowledge: The exact sensor set in your current project is not established; discuss only public or authorised details.

Cross-questions and support: Why fuse sensors? Their strengths and error modes differ. What about slip? Wheel motion may not equal ground motion. What about bias? Inertial errors can accumulate through integration.

Sources: [CONTROL], [ROBOT].

F050 | Explain ROS nodes, topics, services and actions.

Factual answer / explanation: In ROS 2, nodes exchange data through defined interfaces. Topics suit continuous streams using publish/subscribe. Services suit short request/response operations. Actions suit longer-running goals with feedback and cancellation/preemption semantics. These are communication patterns, not guarantees that an application is safe.

Cross-questions and support: Which for GNSS data? A topic is a natural example. Which for a short configuration query? A service. Which for executing a navigation goal? An action can fit. Did your internship use ROS 2? The certificate says ROS only; confirm the version.

Sources: [ROS], [CERT].

F051 | Why do units, timestamps and coordinate frames matter?

Factual answer / explanation: A position or command is meaningful only with its reference frame, units and time. A sensor may be mounted away from the vehicle reference point; stale data or a frame mismatch can create systematic errors. This is a general engineering requirement for interpreting robot state and control inputs.

Supporting knowledge: Suggested checks: explicit units, documented frame transforms, sensor offsets and data-age handling. These are recommendations, not a description of your company architecture.

Cross-questions and support: Latitude/longitude versus metres? They are not interchangeable. Degrees versus radians? Trigonometric calculations must use the intended unit. Can a timestamp mismatch look like steering error? Yes, because measurements may describe a different vehicle state.

Sources: [ROS], [ROBOT], [CONTROL].

F052 | Is an autonomous tractor necessarily an AI system?

Factual answer / explanation: Autonomy can combine deterministic geometry, optimisation, feedback control and learned models. A geometric planner is not automatically machine learning. Your AI module and current algorithm role should be discussed with clear boundaries between learned inference and explicitly programmed logic.

Supporting knowledge: Your certificate 21 establishes a module result, not deployment of a trained model.

Cross-questions and support: When might ML help? Suggested examples: perception or classification where training data is available. When might deterministic methods help? Explicit geometric constraints and reproducible planning. Does AI remove the need for testing? No.

Sources: [CERT], [ASTAR], [ROBOT].

F053 | What are overfitting, data leakage and a fair test?

Factual answer / explanation: As study concepts, overfitting means learning patterns that do not generalise; data leakage means information unavailable at real inference improperly enters training or evaluation. A fair test keeps evaluation independent of model development and matches the intended use. These are general ML explanations, not claims about your course syllabus or employer data.

Supporting knowledge: Use the IIRS workshop as a route to explain what you actually learned, not as proof of ML expertise.

Cross-questions and support: Why split by field or season? Suggested reason: nearby samples can be highly related, so random row splitting may exaggerate generalisation. What should you report? Dataset scope, baseline, errors and limitations, not accuracy alone.

Sources: [CERT].

F054 | What is IIRS and what did your course establish?

Factual answer / explanation: The Indian Institute of Remote Sensing is an ISRO institute at Dehradun focused on capacity building in remote sensing and geoinformatics. Your merit certificate covers Basics of Remote Sensing, GIS and GNSS, 52 hours, with grade B and 73% attendance. Your other IIRS records are participation in shorter programmes.

Cross-questions and support: Is IIRS the same as IARI? No: remote sensing/space versus agricultural research institutions. Is attendance the exam mark? No. Does a participation certificate establish a passed examination? Not by itself.

Sources: [IIRS], [CERT].

F055 | What is NDVI and what can it tell a farmer?

Factual answer / explanation: NDVI=(NIR-Red)/(NIR+Red), using near-infrared and red reflectance. NASA describes it as an index for assessing vegetation greenness/condition, with values from -1 to 1. It can support monitoring, but should not be treated as a direct, unique diagnosis of a particular crop disease or exact yield.

Cross-questions and support: Illustrative calculation? NIR=0.6 and Red=0.2 gives NDVI=0.5. Why not diagnose from one number? Different stresses and background conditions can affect the signal. What next? Ground observations and additional data are needed for interpretation.

Sources: [NDVI].

F056 | How do remote sensing, GIS and GNSS work together?

Factual answer / explanation: Remote sensing provides observations from a distance; GIS organises and analyses spatial information; GNSS provides position/time information. In a suggested farm workflow, imagery may identify patterns, GIS relates them to field boundaries and GNSS helps locate observations or guide machinery.

Supporting knowledge: Your 2024 course explicitly combines these three areas; explain their distinct roles.

Cross-questions and support: Can satellite imagery alone define a safe tractor boundary? Not necessarily: resolution, timing, registration and ground conditions matter. What is ground truth? Independent field observations used to evaluate interpretation.

Sources: [IIRS], [NAV], [CERT].

F057 | What are spatial, spectral and temporal resolution?

Factual answer / explanation: As remote-sensing study concepts: spatial resolution concerns ground detail represented by a measurement; spectral resolution concerns wavelength-band discrimination; temporal resolution concerns observation frequency. They answer different questions and cannot be reduced to one image-quality number.

Supporting knowledge: NRSC’s OCM products illustrate separate spatial, temporal and spectral characteristics; do not generalise one sensor’s specification to all satellites.

Cross-questions and support: Can a coarse image identify a narrow field edge reliably? It may not. Does more frequent imaging guarantee cloud-free optical data? No. What is radiometric resolution? The ability to distinguish signal levels.

Sources: [IIRS], [NISAR].

F058 | What is fNIRS, and what does it not do?

Factual answer / explanation: Functional near-infrared spectroscopy is a non-invasive optical method that estimates changes in oxygenated and deoxygenated haemoglobin in tissue and thereby indirectly studies brain activity. It does not directly read thoughts or establish a person’s intelligence, intentions or workplace ability.

Supporting knowledge: Your certificate is assistance at a workshop, not a clinical or neuroscience qualification.

Cross-questions and support: Is it the same as EEG? No: the physical measurement differs; fNIRS concerns haemodynamic optical signals. Does workshop assistance mean you operated the instrument? Only your actual tasks can establish that.

Sources: [FNIRS], [CERT].

F059 | What is the difference between patent, copyright and an idea?

Factual answer / explanation: At a high level, a patent protects a qualifying technical invention after examination and grant; copyright protects original expression such as code or writing, not the abstract idea alone. Your workshop attendance does not establish either a granted patent or a registered work.

Supporting knowledge: This is interview-level orientation, not legal advice on ownership or patentability.

Cross-questions and support: Can every algorithm be patented? Do not make that claim; patentability depends on applicable law and the claimed invention. Can you publish employer code? Not without the relevant rights and authorisation. What should you say about your workshop? Explain the distinction and what you learned.

Sources: [IP], [CERT].

14 | Category F: mathematics and whiteboard questions

F060 | How do you find the distance between two planar points?

Factual answer / explanation: For points (x1,y1) and (x2,y2), Euclidean distance is sqrt((x2-x1)^2+(y2-y1)^2). It follows from the Pythagorean theorem. It is appropriate when coordinates are in a consistent Cartesian metric frame, not when raw latitude/longitude degrees are treated as metres.

Supporting knowledge: This is a mathematical derivation; no conversion of your source coordinates is performed.

Cross-questions and support: Illustrative example? From (0,0) to (3,4), distance is 5 units. Why not use degree differences directly? Longitude scale varies with latitude and the Earth is curved. What would you do for geodetic coordinates? Use a justified geodesic calculation or a suitable local projection.

Sources: [ROBOT].

F061 | What is curvature and how is it related to radius?

Factual answer / explanation: For a circle, curvature magnitude is 1/R. For a planar curve parameterised by arc length s, curvature describes how quickly heading changes with distance. In the ideal bicycle model, curvature=tan(delta)/L. Smaller radius means larger curvature and more demanding steering.

Cross-questions and support: What is a straight line’s curvature? Zero. What is the curvature of a 5 m radius circle? 0.2 per metre. Does continuous position imply continuous curvature? No: joined segments can have abrupt direction or curvature changes.

Sources: [ROBOT].

F062 | What are dot and cross products useful for in geometry?

Factual answer / explanation: For planar vectors a and b, dot product a_x*b_x+a_y*b_y relates to angle and projection. The scalar 2D cross expression a_x*b_y-a_y*b_x indicates signed orientation under a stated coordinate convention. These are useful for turn direction, side tests and geometric reasoning.

Supporting knowledge: Illustrative example: a=(1,0), b=(0,1) gives dot=0 and cross=1 in a standard x-right/y-up frame.

Cross-questions and support: What does zero cross expression mean? Collinearity in exact arithmetic. What about floating point? Use a justified tolerance. Does a positive sign always mean left? Only after fixing axes and ordering; screen coordinates may reverse visual intuition.

Sources: [ROBOT].

F063 | Why can averaging headings fail?

Factual answer / explanation: Angles wrap around. The arithmetic average of 359 degrees and 1 degree is 180 degrees, which is wrong for their nearby directions. A circular mean uses summed sine and cosine components and atan2, with care when the resultant is near zero.

Supporting knowledge: This is a worked mathematical explanation, not a claim about your code.

Cross-questions and support: What is the correct nearby mean? Zero degrees for equally weighted 359 and 1. What about 0 and 180? Their mean direction is ambiguous because the resultant cancels. Why use atan2? It preserves quadrant information.

Sources: [ROBOT].

F064 | What is numerical tolerance and why does it matter?

Factual answer / explanation: Floating-point calculations approximate real arithmetic. Geometric predicates near tangency, collinearity or a boundary can be sensitive to rounding. A tolerance should relate to units, scale and the application’s accuracy requirements, not be copied as an unexplained constant.

Supporting knowledge: Suggested answer: separate numerical tolerance from physical clearance margin.

Cross-questions and support: Can one epsilon work for degrees and metres? Not meaningfully without conversion and context. What should tests include? Near-boundary and near-degenerate cases. Does increasing tolerance always improve safety? No: it can incorrectly accept invalid geometry.

Sources: [ROBOT], [ASTAR].

F065 | What is the difference between interpolation and extrapolation?

Factual answer / explanation: Interpolation estimates between known samples; extrapolation extends beyond their range. A fitted headland-width formula can look accurate within measured conditions yet fail outside them. A mathematical fit is not automatically a physical law.

Supporting knowledge: Use only actual measured data when discussing your work; the handbook does not invent a fitted company formula.

Cross-questions and support: How would you validate a polynomial? Hold out cases, inspect residuals, test boundary conditions and state the valid domain. Why can a high-degree fit be risky? It may oscillate or behave poorly outside the sample range.

Sources: [ICAR].

F066 | How do you explain algorithm complexity?

Factual answer / explanation: Complexity describes how resource use grows with input size under stated assumptions. For graph search, nodes, edges, representation and data structures matter. A theoretical bound and measured runtime answer different questions.

Supporting knowledge: Do not quote one universal A* complexity without specifying the graph and implementation.

Cross-questions and support: Does O(n) mean one second? No. Can a faster average hide a dangerous worst case? Yes. What should a real-time discussion include? Deadline, hardware, input bounds, timeout behaviour and measured worst-case evidence.

Sources: [ASTAR].

F067 | How do you calculate percentage improvement correctly?

Factual answer / explanation: For a quantity to minimise, relative reduction=(baseline-new)/baseline*100%. State the baseline, units and whether the comparison is paired under equivalent conditions. A percentage-point change is different from a relative percentage change.

Supporting knowledge: These are hypothetical arithmetic examples, not your project results.

Cross-questions and support: Illustrative example? Reducing 100 m to 90 m is a 10% reduction. Raising coverage from 90% to 95% is 5 percentage points, or about 5.56% relative increase. Can this prove safety? No: it measures only the defined metric.

Sources: [FARM].

15 | Category F: mountaineering, climbing and Arunachal

F068 | What is NIMAS and where is it?

Factual answer / explanation: NIMAS is the National Institute of Mountaineering and Adventure Sports at Dirang in West Kameng district, Arunachal Pradesh. Your certificate records completion of its Basic Mountaineering Course and recommendation for an advanced course.

Supporting knowledge: The certificate’s grade and recommendation do not establish every expedition or peak you have climbed.

Cross-questions and support: Is NIMAS the same as NIM Uttarkashi or HMI Darjeeling? No: they are different institutes. What did your course include? Use your actual course log; public BMC syllabi provide context but do not prove your exact daily training.

Sources: [NIMAS], [BMC], [HMI], [CERT].

F069 | How do trekking, mountaineering and sport climbing differ?

Factual answer / explanation: At an introductory level, trekking is travel on foot through terrain; mountaineering can involve technical mountain travel on rock, snow or ice; sport climbing focuses on climbing routes under specific protection and competition/recreation systems. They overlap in fitness and movement skills but require different technical training.

Supporting knowledge: Prepare the techniques actually practised under instruction, without offering to teach safety-critical systems beyond your qualification.

Cross-questions and support: Is a trek certificate equivalent to BMC? No. Is an indoor climbing grade a mountain-altitude qualification? No. What did you complete? BMC and sport climbing Level 1 as documented, not an advanced course.

Sources: [BMC], [HMI], [CERT].

F070 | Why is high altitude difficult even for a fit runner?

Factual answer / explanation: CDC explains that lower atmospheric pressure reduces the partial pressure of oxygen. Fitness does not remove the need for acclimatisation. Acute acclimatisation involves increased ventilation; increased red-cell production is not the immediate mechanism in the first few days.

Supporting knowledge: This is educational preparation, not personal medical clearance.

Cross-questions and support: Does oxygen percentage simply disappear? The key issue is reduced pressure and inspired oxygen partial pressure. Can you predict altitude tolerance from a 10 km time? No. Why sleep altitude matters? Hypoxaemia can be greatest during sleep.

Sources: [ALT].

F071 | What are AMS, HACE and HAPE?

Factual answer / explanation: AMS is acute mountain sickness, commonly involving headache with other symptoms after ascent. HACE is high-altitude cerebral oedema and HAPE high-altitude pulmonary oedema; both can be life-threatening. CDC highlights confusion/loss of coordination for HACE and serious breathing problems for HAPE.

Supporting knowledge: Do not turn this handbook into a rescue manual or substitute it for instructor-led training and medical advice.

Cross-questions and support: What is the safe interview principle? Do not ascend with symptoms; worsening symptoms require descent and medical help. What about severe symptoms? Treat as an emergency, seek rescue/medical support and descend as safely as possible. Should you prescribe medication? No: medication decisions belong to qualified clinicians.

Sources: [ALT], [ALT2].

F072 | Why are dynamic and low-stretch ropes not interchangeable?

Factual answer / explanation: Dynamic climbing ropes are designed and tested to absorb fall energy through elongation; UIAA 101 is the dynamic-rope standard. Low-stretch/static ropes serve different controlled-use requirements. The correct rope depends on the activity, equipment system and manufacturer/instructor guidance.

Supporting knowledge: No step-by-step rigging or knot instructions are provided here because this is interview preparation, not practical certification.

Cross-questions and support: Can you select a rope just by colour? No. Does a certification mark remove the need for inspection? No. Should a basic-course graduate improvise a rescue rig? No: follow trained procedures and qualified supervision.

Sources: [ROPE], [BMC].

F073 | What safety knowledge should you revise from your actual course?

Factual answer / explanation: Revise the purpose and limitations of helmet, harness, carabiners, ropes, belay devices, crampons and ice axe only to the level actually taught. Be able to explain inspection, partner checks, instructor commands and why equipment compatibility matters.

Supporting knowledge: Use your course manual and instructor guidance for practical technique; this card is a revision checklist.

Cross-questions and support: Which knots did you learn? Supply your real course list and purpose, not a memorised list you cannot demonstrate safely. What is a turnaround decision? A pre-agreed safety decision based on conditions and group capability, not failure of ambition.

Sources: [BMC], [HMI], [CERT].

F074 | What should you know about Arunachal Pradesh?

Factual answer / explanation: Its capital is Itanagar. It borders Bhutan, China/Tibet and Myanmar internationally, and Assam and Nagaland within India. Dirang is in West Kameng. The official state portal gives area 83,743 square kilometres and identifies it as the Land of Dawn-lit Mountains.

Supporting knowledge: Do not quote changing district counts or disputed boundary lengths without a dated authoritative source.

Cross-questions and support: Why is it strategically important? Suggested analysis: international borders, mountain terrain, connectivity and logistics. What river valleys should you locate? Kameng, Subansiri, Siang, Lohit and Tirap are listed in official profiles. What did you personally observe? Give actual travel experience, not stereotypes.

Sources: [ARUN], [ARUN2], [CERT].

F075 | How can mountain experience relate to an officer interview without exaggeration?

Factual answer / explanation: The factual basis is your completed course. The personal connection should come from actual examples of preparation, following safety instructions, sharing loads, managing limited resources and making conservative decisions. Adventure participation is not proof of military leadership or operational readiness.

Cross-questions and support: What if the interviewer says a course is only recreation? Explain what you actually did and learned, without claiming equivalence to military training. What is the strongest evidence? A specific action and outcome rather than the grade alone.

Sources: [CERT].

16 | Category F: running, cricket, volleyball and BSF

F076 | What were your recorded running paces?

Factual answer / explanation: The 10 km certificate gives 53:32, equivalent to 321.2 seconds/km, or about 5:21.2 per km, and average speed about 11.21 km/h. The Faridabad certificate prints 21 KM and 2:17:01; using that printed distance gives about 6:31.5 per km and 9.20 km/h.

Supporting knowledge: The calculations use certificate values as printed. A half-marathon event label does not itself certify the exact measured course length.

Cross-questions and support: Why not calculate using 21.1 km? Because the source prints 21 KM; do not silently replace it with a standard race distance. What if the official race result confirms another distance? State that separate source and recalculate explicitly. Can pace difference establish why you slowed? No: conditions and preparation require your account.

Sources: [CERT].

F077 | What is net/chip time versus gun time?

Factual answer / explanation: As a race-timing concept, gun time runs from the official start signal; net/chip time runs from an individual’s start-line crossing to finish-line crossing. Your Bordermen certificate explicitly says chip time and the Faridabad certificate says net time.

Cross-questions and support: Which is used for awards? It depends on event rules; do not assume. Does position 287 reveal percentile? No: the denominator and category are absent. Did you run a marathon at the BSF event? Your category was 10 km, regardless of the event’s name.

Sources: [CERT].

F078 | What is BSF and when was it raised?

Factual answer / explanation: The Border Security Force was raised on 1 December 1965. Its official history links its creation to the need for a centrally controlled, equipped border force after the April 1965 attacks in Kutch. K. F. Rustamji was its first Director General.

Cross-questions and support: Is BSF the Indian Army? No: distinguish the border-guarding force from the Army. Why is it relevant to your PIQ? You participated in its Amritsar-to-Attari event. What do you know personally? Event organisation, route and your category, not operational details.

Sources: [BSF], [CERT].

F079 | What basic cricket dimensions and governance should you know?

Factual answer / explanation: MCC Law 6 defines the pitch as 22 yards/20.12 m long and 10 feet/3.05 m wide. MCC is custodian of the Laws; ICC is the global governing body and applies playing conditions at international level. Laws and competition-specific playing conditions are not identical.

Supporting knowledge: Prepare your actual playing role, bowling/batting style and school event details separately.

Cross-questions and support: Does every cricket ground have the same boundary? No: do not confuse pitch dimensions with whole-ground dimensions. Which law edition? MCC’s site lists a 2026 fourth edition as well as earlier editions; check the applicable competition conditions before quoting a recent rule.

Sources: [CRIC], [CRIC2].

F080 | What should a volleyball player know immediately?

Factual answer / explanation: Under FIVB indoor rules, the court is 18 m by 9 m. Six players per team are on court. A match is won by three sets; ordinary sets go to 25 points and a deciding fifth set to 15, with a two-point lead required. A team normally has up to three hits in addition to a block contact.

Supporting knowledge: Use the current 2025-2028 FIVB rules and your event’s adopted format; local short matches may use modified formats.

Cross-questions and support: What happens on winning the right to serve? Players rotate according to the rules. Is every net touch automatically the same fault? Apply the actual rule and context, not a slogan. What is a libero? A specialised defensive player with defined restrictions.

Sources: [VOL].

F081 | What are the volleyball net heights and key positional ideas?

Factual answer / explanation: FIVB specifies 2.43 m for the men’s net and 2.24 m for the women’s net. The attack line is 3 m from the centre line. Front-row/back-row status and rotation affect legal actions; the setter, attackers and defensive specialists have tactical roles but must follow the same rotation framework.

Supporting knowledge: Your department event may have used a simplified format; state that rather than claiming all local rules were FIVB-standard.

Cross-questions and support: Can a back-row player attack freely from anywhere? No: restrictions apply, particularly relative to the attack line and ball height. Does a block count as one of the three team hits? A block contact is treated separately under the indoor rules.

Sources: [VOL].

17 | Category F: NSS, civic responsibility and selection awareness

F082 | What is NSS and what is its motto?

Factual answer / explanation: NSS is the National Service Scheme under the Ministry of Youth Affairs and Sports. It was launched on 24 September 1969. Its motto is Not Me But You. The official FAQ describes two years of service, 120 hours each year, and a seven-day special camp.

Cross-questions and support: Does your certificate match that framework? It records 240 hours across 2022-23 and 2023-24 and a seven-day camp. Is NSS the same as NCC? No: do not claim military training from NSS. What is its aim? Education and personal development through community service.

Sources: [NSS], [CERT].

F083 | How do you measure whether a service activity helped?

Factual answer / explanation: Suggested framework: identify the community need, define your task, record participation/output, seek feedback and check follow-up. Attendance is an output; a lasting improvement is an outcome and requires evidence. Avoid claiming that a camp solved a problem without data.

Supporting knowledge: NSS emphasises understanding community needs and involving people in problem solving.

Cross-questions and support: Medical camp example? You may count registrations or logistics completed if recorded, but cannot infer health improvement from attendance alone. What if a programme was poorly attended? Explain your own corrective action, not blame.

Sources: [NSS].

F084 | Which constitutional duties connect to your profile?

Factual answer / explanation: Article 51A includes respect for the Constitution and institutions, protecting sovereignty/unity/integrity, national service when called upon, harmony, environmental protection, scientific temper and striving for excellence. Your engineering, service and environmental examples can illustrate your own understanding.

Supporting knowledge: This is civic interview preparation, not legal advice or a claim about a particular dispute.

Cross-questions and support: Is scientific temper only for scientists? No: it concerns inquiry and evidence-based reasoning. Can patriotism coexist with respectful disagreement? Your answer should recognise constitutional institutions and lawful democratic discussion.

Sources: [DUTY].

F085 | What are the stages of SSB and what does recommendation mean?

Factual answer / explanation: The Army’s overview distinguishes the broader selection process from the two stages within SSB. SSB screening includes intelligence and picture-perception/discussion testing. Later assessment uses interviewing, group-testing and psychological techniques, followed by conference. Recommendation is followed by medical and other eligibility/merit requirements; it is not a joining guarantee.

Supporting knowledge: Older generic instructions and newer overviews differ in some details; do not treat an old checklist as your current call-up order.

Cross-questions and support: Are the written exam and SSB screening the same stage? No: be clear which framework you mean. Does being medically fit guarantee a vacancy? No. Which instructions control your reporting? Your current entry-specific notification and call-up letter.

Sources: [SEL], [SSB].

F086 | Why should you verify entry-specific terms rather than memorise a coaching page?

Factual answer / explanation: The Army overview explicitly says the notification for the specific entry and course governs the selection process. Course numbers can overlap across different entry schemes. Your PIQ says UPSC SSC(NT) Men-125; a search result for NCC Special Entry 125 is not proof of your terms.

Supporting knowledge: This is an intentional evidence gap, not an invitation to substitute a plausible but different course.

Cross-questions and support: What is not fully verified here? Your exact course notification, academy joining date, training duration, service-tenure clauses and current document checklist. What should you use? The actual notification and call-up letter you received.

Sources: [SEL], [PIQ].

18 | Category F: military fundamentals and public defence knowledge

F087 | Name the commissioned Army ranks in order.

Factual answer / explanation: From junior to senior: Lieutenant, Captain, Major, Lieutenant Colonel, Colonel, Brigadier, Major General, Lieutenant General and General. Field Marshal is the exceptional five-star rank above General, not the routine next promotion for every officer.

Supporting knowledge: Study official rank-badge images separately; this handbook does not reproduce insignia from memory.

Cross-questions and support: Army Lieutenant equivalent? Navy Sub Lieutenant and Air Force Flying Officer. Army Colonel equivalent? Navy Captain and Air Force Group Captain. Why is Captain confusing? Navy Captain is not equivalent to Army Captain.

Sources: [RANK].

F088 | What are the Army commands and headquarters?

Factual answer / explanation: The standard seven-command list is Northern-Udhampur, Western-Chandimandir, South Western-Jaipur, Southern-Pune, Eastern-Kolkata, Central-Lucknow and Army Training Command-Shimla. Six are operational commands and one is the training command.

Supporting knowledge: Some coaching pages incorrectly say seven operational plus one training command; do not repeat that count. Headquarters and organisational changes should still be rechecked before interview.

Cross-questions and support: Which is in Uttar Pradesh? Central Command at Lucknow. Is Central Command only a rear-area logistics organisation? No: its official page describes operational responsibility in the central sector. Is Mathura’s corps assignment fixed forever? No; use dated current information.

Sources: [COMMAND], [CENTRAL], [CORPS26].

F089 | What is the difference between rank and appointment?

Factual answer / explanation: Rank is a person’s military grade; appointment is the position held. General Officer Commanding is an appointment description, while Lieutenant General is a rank. The same rank can hold different appointments.

Cross-questions and support: Is CDS a separate military rank? It is an appointment held by a senior officer. Is a corps a regiment? No: distinguish formation levels and branch-specific use of the word regiment. Why avoid exact troop counts? Organisation varies by formation and role.

Sources: [RANK], [CDS], [CORPS].

F090 | What are arms and services, and where might engineering be relevant?

Factual answer / explanation: The Army’s official list includes Infantry, Armoured Corps, Artillery, Army Air Defence, Army Aviation, Engineers, Signals, Mechanised Infantry and supporting services such as ASC, AOC, EME and AMC. Public overviews describe Engineers in mobility/counter-mobility, Signals in communications and EME in equipment maintenance/repair.

Supporting knowledge: No arm or service is ranked as more valuable here; operational effectiveness depends on combined functions.

Cross-questions and support: Engineers versus EME? Their broad roles differ; do not assume every engineering graduate joins either. ASC versus AOC? The public overview distinguishes supplies/transport

from ordnance/equipment support. What is your preference? That remains a personal answer, not a guaranteed allocation.

Sources: [ARMS], [ARMS2].

F091 | What is the CDS role, and how is it different from a service chief?

Factual answer / explanation: The official IDS biography describes the CDS as principal military adviser to the Defence Minister on tri-service matters, permanent chairman of the Chiefs of Staff Committee and Secretary, Department of Military Affairs. A service chief leads the respective service. Do not describe the CDS as simply replacing all service chiefs.

Cross-questions and support: What is jointness? Suggested explanation: services coordinating capabilities, planning and support toward shared objectives. What is integration? It can involve more enduring organisational and functional arrangements. Is every proposed theatre reform already implemented? No: distinguish proposals from notified structures.

Sources: [CDS].

F092 | What military history should you prepare without confusing operation names?

Factual answer / explanation: The government’s published official histories include Jammu and Kashmir operations 1947-48, Operation Polo 1948, Operation Vijay for Goa in 1961, the 1965 India-Pakistan war and the 1971 war. Use these as anchors for further study of causes, geography, major outcomes and lessons, rather than memorising slogans.

Supporting knowledge: The 1962 war and 1999 Kargil conflict require their own reliable historical sources; no detailed battle account is invented here.

Cross-questions and support: Can an operation name be reused? Yes; specify year and context. Does this list cover every war? No: it is a list of published histories, not a complete conflict chronology. What should you know personally? One well-researched battle or military leader relevant to your interest, without fabricated anecdotes.

Sources: [HISTORY].

F093 | What is Operation Sindoor in a carefully dated answer?

Factual answer / explanation: The initial Ministry of Defence release on 7 May 2025 states that the armed forces struck nine sites of terrorist infrastructure in Pakistan and Pakistan-occupied Jammu and Kashmir following the Pahalgam attack. It described the initial action as focused, measured and non-escalatory and said Pakistani military facilities were not targeted in that initial action.

Cross-questions and support: Why specify initial? Later phases and responses require their own dated sources; do not generalise one release to the whole sequence. What lesson relates to your work? As analysis, reliable information, coordination and tested technology matter. What should you avoid? Unverified casualty or aircraft-loss claims.

Sources: [SINDOOR].

F094 | What is the difference between a carrier, a submarine and a torpedo?

Factual answer / explanation: A carrier provides a sea-based platform for aircraft operations. A submarine operates underwater. A torpedo is an underwater weapon; it is not the same as a surface-to-air or cruise missile. The Navy’s public carrier account identifies INS Vikramaditya and the indigenously built INS Vikrant in combined operations.

Cross-questions and support: Is a carrier only a large transport ship? No: its aviation role is central. Is Varunastra a submarine? No: the current contract describes it as a ship-launched anti-submarine torpedo.

Can you quote current patrol locations? Do not speculate about operational deployments.

Sources: [CARRIER], [DEAL2].

19 | Current affairs: verified snapshot, 16 October 2026

Study the dated facts first. Then develop your own balanced view. Forecasts are not outcomes, contracts are not deliveries, and a trial is not proof of operational deployment. Refresh this section before your actual interview.

F095 | Who are the current CDS and service chiefs?

Factual answer / explanation: As checked for this edition: CDS General N. S. Raja Subramani, in office from 31 May 2026; Army Chief General Dhiraj Seth, effective 30 June 2026; Navy Chief Admiral Krishna Swaminathan, from 31 May 2026; Air Chief Marshal A. P. Singh is listed as Chief of the Air Staff on the IAF portal.

Supporting knowledge: Recheck official portals before interview; do not rely on a 2025 chiefs list.

Cross-questions and support: Why might older notes differ? Several appointments changed in 2026. What should you remember beyond names? The role of each appointment. Who is Defence Minister? The MoD directory lists Rajnath Singh.

Sources: [CDS], [ARMY], [NAVY], [IAF], [MOD].

F096 | What defence contracts were announced on 15 October 2026?

Factual answer / explanation: Akashvani reports contracts totalling over Rs 4,000 crore for GSAT-7C communication-satellite capability for the IAF and Varunastra heavyweight torpedoes for the Navy. It gives over Rs 2,500 crore for GSAT-7C with NSIL and over Rs 1,500 crore for Varunastra with BDL.

Supporting knowledge: This is directly relevant to your navigation, communication and systems-engineering interests, but no classified capability is discussed.

Cross-questions and support: Contract versus induction? Signing does not mean delivery or operational deployment. Who developed Varunastra? The report identifies NSTL. What does NSIL do here? It is the contracted entity for the satellite capability; do not confuse it with a service headquarters.

Sources: [DEAL2].

F097 | Why is GSAT-7C different from NavIC?

Factual answer / explanation: The 15 October report describes GSAT-7C as a dedicated communication capability for the IAF. NavIC is a positioning/navigation/timing system. Communication transfers information; navigation estimates position/time. A vehicle may need both, but one is not a substitute for the other.

Cross-questions and support: Does satellite communication guarantee a GNSS fix? No. Does GNSS provide an internet link? Not by itself. What happens when either is unavailable? The system needs a validated degraded-mode policy; do not invent your employer’s thresholds.

Sources: [DEAL2], [NAV].

F098 | What recent ADRDE development connects directly to Agra?

Factual answer / explanation: The MoD release dated 6 October 2026 reports a flight trial of an indigenous lighter-than-air High-Altitude Platform developed by ADRDE, Agra. It reached 21 km above mean sea level and remained around 20 km for more than 30 minutes before commanded descent and recovery. The release lists IMU, GPS receiver, cameras and altitude-control instrumentation.

Supporting knowledge: Use the primary MoD release rather than repeating speculative endurance or deployment claims.

Cross-questions and support: Is it a satellite? No: it is an atmospheric platform. Does a 30-minute trial establish months of endurance? No. How does it connect to your work? General sensing, control, telemetry and validation concepts, not identical design requirements.

Sources: [HAPS0].

F099 | What is the FY2026-27 defence budget headline?

Factual answer / explanation: PIB's 3 February backgrounder gives a Ministry of Defence allocation of Rs 7.85 lakh crore, a 15.19% increase over FY2025-26 Budget Estimates. It identifies over Rs 2.19 lakh crore under the capital head and Rs 1.39 lakh crore earmarked for procurement from domestic defence industries.

Supporting knowledge: The backgrounder gives DRDO allocation Rs 29,100.25 crore. Retain the fiscal year and budget basis when quoting it.

Cross-questions and support: Budget estimate versus actual spending? An allocation is not proof of expenditure or delivery. Capital versus revenue? Broadly, capital supports asset creation/acquisition while revenue supports ongoing requirements; use the budget's exact categories. Why compare BE with BE? Mixing BE and revised/actual figures can mislead.

Sources: [BUDGET].

F100 | What did RBI decide on 7 October 2026?

Factual answer / explanation: RBI's MPC resolution raised the policy repo rate by 25 basis points to 5.50%. SDF became 5.25% and MSF/Bank Rate 5.75%. The stance changed to calibrated tightening. The meeting was chaired by Governor Sanjay Malhotra.

Supporting knowledge: RBI projected FY2026-27 real GDP growth at 7.1% and CPI inflation at 5.2%; these are projections, not final outcomes.

Cross-questions and support: What is 25 basis points? 0.25 percentage point. Why not call it a 25% rise? That is a different calculation. Does the policy rate equal every tractor-loan rate? No: lending rates also depend on transmission, product and borrower terms.

Sources: [RBI].

F101 | How could interest rates and inflation affect tractors and farmers?

Factual answer / explanation: As analysis, borrowing costs can affect financed purchases and investment; fuel and input-price pressures can affect operating costs; rural income and weather can affect demand. RBI's October resolution specifically discusses energy prices, deficient monsoon/El Nino risks and rural demand.

Supporting knowledge: Keep this as an analytical chain, not a forecast for your employer.

Cross-questions and support: Would a rate rise always reduce tractor sales immediately? No: demand has multiple drivers and transmission takes time. How would you test the relationship? Use dated sales, credit, crop-income and weather data rather than one anecdote.

Sources: [RBI], [EKL].

F102 | What does the World Bank’s October India update say?

Factual answer / explanation: The World Bank’s 6 October 2026 India Development Update projects 7.1% growth for FY27, supported by domestic demand and exports. It highlights risks including oil prices, El Nino and capital-flow volatility. Its special focus discusses AI opportunities and the need for skills, infrastructure and broad access.

Cross-questions and support: Is 7.1% an actual measured annual result? No: it is a forecast. Why is AI relevant to agriculture? Suggested examples include locally adapted decision support and automation, with access and data-quality constraints. Does AI guarantee job creation? The report says long-term labour-market impacts remain uncertain.

Sources: [WB].

F103 | What is the global economic picture in the October IMF outlook?

Factual answer / explanation: The October 2026 IMF WEO projects global growth of 3.1% in 2026 and 3.4% in 2027, with global headline inflation 4.7% in 2026 and 4.1% in 2027. It identifies conflict, trade tensions, fiscal vulnerabilities and AI-related expectations as important risks.

Cross-questions and support: Why distinguish global and Indian fiscal-year figures? They refer to different populations and potentially different reporting periods. Can a global forecast prove India’s exact growth? No. What is a useful interview view? Explain transmission through energy, trade, investment and financial conditions.

Sources: [IMF].

F104 | What happened at the 2026 BRICS summit?

Factual answer / explanation: The official New Delhi Declaration records the eighteenth BRICS summit in New Delhi on 12-13 September 2026 under the theme Building for Resilience, Innovation, Cooperation and Sustainability. The official portal lists eleven members, including Indonesia, and ten partner countries.

Supporting knowledge: For a membership question, use the official list in the next card rather than an old five-member mnemonic.

Cross-questions and support: Is BRICS a military alliance? The official description is a cooperation/consultation platform, not a collective-defence pact. Why should India engage? Develop your view around development finance, reform of institutions, trade and strategic autonomy. Are members identical in interests? No: cooperation does not remove differences.

Sources: [BRICS], [BRICSD].

F105 | Name the BRICS members and distinguish partners.

Factual answer / explanation: The official India 2026 portal lists Brazil, China, Egypt, Ethiopia, India, Indonesia, Iran, Russia, Saudi Arabia, South Africa and the UAE as members. It lists Belarus, Bolivia, Cuba, Kazakhstan, Malaysia, Nigeria, Thailand, Uganda, Uzbekistan and Vietnam as partners.

Supporting knowledge: Membership descriptions can differ across older sources; this card follows the official India chairship portal as accessed for this edition.

Cross-questions and support: When did Indonesia join? The portal says 2025. How many times has India chaired? Four, including 2026; earlier 2012, 2016 and 2021. Are partner and member interchangeable? No.

Sources: [BRICS].

F106 | What is the current India-China diplomatic position in a sourced answer?

Factual answer / explanation: The PMO account of the 12 September 2026 meeting says the leaders reviewed relations and that India emphasised peace and tranquillity in border areas, observance of agreements and a fair, reasonable, mutually acceptable boundary resolution. It also mentions structural trade imbalance, supply chains and market access.

Supporting knowledge: Avoid unsupported claims about current troop positions or tactical incidents.

Cross-questions and support: Does a summit mean the boundary dispute is settled? No. Can trade and security both matter? Yes: explain the trade-off without claiming either disappears. Why is this relevant to you? Arunachal travel, mountain logistics and technology supply chains provide personal study links.

Sources: [CHINA], [ARUN].

F107 | Why does West Asia matter to India?

Factual answer / explanation: RBI and IMF identify conflict-related energy and supply-chain risks. Akashvani’s March advisory report describes travel disruption and consular safety measures. An interview answer can connect these verified themes to energy security, shipping, Indian citizens abroad and economic stability.

Cross-questions and support: What is your policy view? Personal answer: prioritise civilian safety, diplomacy, secure supply routes and verified information. Can you quote a current evacuation total? Not from these sources; no current total is asserted. Is an old advisory automatically valid today? No: check the latest mission guidance.

Sources: [RBI], [IMF], [WEST].

F108 | What were India’s 2026 Asian Games results?

Factual answer / explanation: Olympics.com reports 85 medals: 21 gold, 27 silver and 37 bronze, placing India fourth at Aichi-Nagoya. It identifies athletics as contributing 24 medals and archery nine, including five golds. The Games ran 19 September-4 October 2026.

Supporting knowledge: Do not confuse the Asian Games with the Asian Para Games or Olympics.

Cross-questions and support: How did that compare with Hangzhou? The same source gives 106 medals including 28 golds for Hangzhou 2023. Does a lower total alone explain the reasons? No: event mix, qualification and performance need evidence. Why relevant? Your running and volleyball interests make sports awareness a natural route.

Sources: [ASIAN].

F109 | What is NISAR and why does it connect to your IIRS learning?

Factual answer / explanation: NRSC describes NISAR as the NASA-ISRO Synthetic Aperture Radar Earth-observation mission, launched on 30 July 2025, using L- and S-band radar to study surface changes. Bhoonidhi provides access to satellite data and lists NISAR S-SAR processed collections from July 2026.

Cross-questions and support: Is radar imagery the same as an optical photograph? No: the sensing physics differs. Can it support agriculture? It can contribute observations, but a crop-specific conclusion needs appropriate processing and validation. Is data availability proof that you used it? No: state only your actual experience.

Sources: [NISAR].

P065 | Which current issue would you choose for a two-minute discussion?

Your answer preparation: Choose one you understand: farm automation affordability, sovereign navigation, indigenous defence R&D, energy security or AI access. State the dated event, why it matters, competing considerations and your reasoned view.

Supporting knowledge: Use the factual cards above. Do not memorise an opinion as if it were a fact.

Cross-questions and support: What is the strongest counterargument? What evidence would change your view? What is your proposed practical step?

Sources: [RBI], [WB], [HAPS0], [BRICS].

P066 | Should India prioritise domestic defence production even if imports are available?

Your answer preparation: Develop your own balanced position. Consider reliability, readiness, lifecycle support, technology control, cost, quality and delivery timelines. Do not assume indigenous automatically means cheaper or immediately available.

Supporting knowledge: The budget earmarks domestic procurement funds; the latest contracts and ADRDE trial provide concrete examples, but do not settle every procurement trade-off.

Cross-questions and support: What if domestic delivery is delayed? What if imported spares become unavailable? How would you judge a hybrid approach?

Sources: [BUDGET], [DEAL2], [HAPS0].

P067 | Will autonomous tractors reduce rural employment?

Your answer preparation: Offer a balanced personal view: tasks can change, labour burden may fall, and new service/maintenance roles may emerge, but distributional effects depend on adoption and local conditions. Avoid claiming guaranteed net job creation or loss without evidence.

Supporting knowledge: FAO includes social and economic considerations in sustainable mechanisation; the World Bank says long-term AI labour impacts remain uncertain.

Cross-questions and support: What about small farms? Who pays for training? What if a machine fails during a narrow sowing window?

Sources: [FAO], [WB].

20 | Category P: situational judgement and pressure questions

These are hypothetical practice situations, not claims about events in your life. Explain your own decision and priorities. The support notes are suggested reasoning, not an official SSB scoring key.

P068 | A demonstration is due, but you discover a boundary-crossing bug.

Your answer preparation: Explain how you would prevent unsafe operation, communicate the limitation, preserve the evidence and propose a safe alternative demonstration.

Supporting knowledge: Suggested support: a simulator demonstration or restricted test can be more honest than presenting an unvalidated fix as safe.

Cross-questions and support: What if the fix is untested? What if a deadline is missed?

Sources: [CERT], [GNSS], [ALT], [FAO].

P069 | GNSS quality degrades while the tractor is close to a field edge.

Your answer preparation: Discuss the system’s authorised degraded-mode policy and your role in it. Do not invent a universal continue/stop threshold.

Supporting knowledge: Supporting concept: accuracy and integrity differ; uncertainty and stopping capability matter, not just smooth coordinates.

Cross-questions and support: What if the displayed position looks stable? What if communications also fail?

Sources: [CERT], [GNSS], [ALT], [FAO].

P070 | Your algorithm gives a shorter route but misses a narrow strip.

Your answer preparation: State the actual task objective and why a distance-only metric is insufficient. Explain how you would compare complete feasible coverage alternatives.

Supporting knowledge: Suggested support: report coverage, overlap, time and violations separately; a hard requirement cannot be traded away silently.

Cross-questions and support: Would you hide the missed strip in an average? What if the user prefers speed?

Sources: [CERT], [GNSS], [ALT], [FAO].

P071 | You find an error in your own submitted result.

Your answer preparation: Explain prompt correction, impact assessment, notification and a preventive check. Keep responsibility specific.

Supporting knowledge: Suggested support: preserve the original and corrected result with clear versioning rather than rewriting the history.

Cross-questions and support: What if nobody has noticed? What if the correction changes the conclusion?

Sources: [CERT], [GNSS], [ALT], [FAO].

P072 | A placement schedule clashes with an examination.

Your answer preparation: Explain how you would confirm constraints, contact the authorised stakeholders, present options and communicate the final decision consistently.

Supporting knowledge: Supporting fact: your coordinator certificate establishes facilitation, not authority to override academic or recruiter decisions.

Cross-questions and support: What if students ask for unequal exceptions? What if a recruiter cannot change time?

Sources: [CERT], [GNSS], [ALT], [FAO].

P073 | An interviewer asks for confidential source code as proof of your work.

Your answer preparation: Explain that you can discuss public concepts and a sanitised independent example, but cannot share protected employer material.

Supporting knowledge: Suggested support: confidentiality can be maintained while demonstrating mathematical understanding and test reasoning.

Cross-questions and support: What if asked for a screenshot? What if the interviewer says everyone shares it?

Sources: [CERT], [GNSS], [ALT], [FAO].

P074 | Your climbing group is close to its objective when conditions worsen.

Your answer preparation: Explain communication with the instructor/leader, adherence to safety limits and willingness to turn back. Do not improvise rescue techniques.

Supporting knowledge: Suggested support: sunk cost does not improve weather or safety; decisions should reflect current conditions and qualified guidance.

Cross-questions and support: What if another participant insists on continuing? What if you have paid a large fee?

Sources: [CERT], [GNSS], [ALT], [FAO].

P075 | A participant develops confusion and poor coordination at altitude.

Your answer preparation: State that this can be a serious warning sign, notify qualified leaders/medical support and prioritise emergency assistance and safe descent.

Supporting knowledge: CDC identifies confusion and loss of coordination as HACE warning signs; practical response must follow trained emergency procedures.

Cross-questions and support: Would you continue to the summit? Would you prescribe medication yourself?

Sources: [CERT], [GNSS], [ALT], [FAO].

P076 | A medical-camp visitor asks you to diagnose a condition.

Your answer preparation: Explain your volunteer role, direct the person to qualified medical staff and protect privacy.

Supporting knowledge: Suggested support: you can assist with logistics and access without acting beyond your competence.

Cross-questions and support: What if the queue is long? What if the person cannot read the form?

Sources: [CERT], [GNSS], [ALT], [FAO].

P077 | A farmer says your autonomous system is too expensive.

Your answer preparation: Listen to the actual task, utilisation and service concerns; discuss measurable value and alternatives without overselling.

Supporting knowledge: Supporting concept: sustainable mechanisation includes economic and social fit; no universal payback period is established.

Cross-questions and support: Would you promise fuel savings? What if a shared-service model is unsuitable?

Sources: [CERT], [GNSS], [ALT], [FAO].

P078 | You are challenged on the essay-result or name discrepancy.

Your answer preparation: State the verified record calmly. If events differ, explain their dates and organisers; if an entry is wrong, acknowledge it and follow the authorised correction process.

Supporting knowledge: Do not invent an affidavit, name-change history or second competition to make the records fit.

Cross-questions and support: Why did you not check earlier? What document supports your explanation?

Sources: [CERT], [GNSS], [ALT], [FAO].

P079 | You do not know the answer to a factual cross-question.

Your answer preparation: Say that you do not know the specific fact, then distinguish any relevant concept you do understand. Avoid guessing a name, date or technical specification.

Supporting knowledge: Suggested support: ask yourself whether the hint is consistent with known facts; respectful uncertainty is better than confident fabrication.

Cross-questions and support: What if the interviewer offers a misleading hint? Would you agree automatically?

Sources: [CERT], [GNSS], [ALT], [FAO].

21 | Connected cross-question chains for mock interviews

Hometown to defence R&D;

Tell me about Sarendhi -> why Agra matters -> cantonment versus city -> ADRDE expansion -> ADRDE versus PTS -> recent HAPS trial -> IMU/GPS roles -> why a trial is not deployment.

Answer support: village details require your records; cantonment established 1805; ADRDE develops aerial-delivery/aerostat systems; PTS is IAF training; 6 October HAPS trial reached 21 km.

Sources: [PIQ], [CANT], [ADR], [PTS], [HAPS0]

Engineering career to Army motivation

Why Agricultural Engineering -> why DEI -> why R&D -> why path planning -> why leave a useful civilian role -> why SSC(NT) -> what if another arm/service -> responsible handover.

Answer support: use personal decisions, not a prewritten motivation. Your exact course notification remains essential; technical background does not guarantee branch allocation.

Sources: [PIQ], [CERT], [SEL]

Algorithm ownership

Describe one algorithm -> inputs/outputs -> constraints -> objective -> baseline -> failure case -> regression tests -> worst case -> what remains outside your responsibility.

Answer support: explain a non-confidential example. Distinguish path length, coverage, tracking and safety; no unsupported improvement percentage.

Sources: [ASTAR], [ROBOT], [CONTROL]

Running to numerical reasoning

Which event -> actual category -> chip time -> pace -> speed -> rank denominator -> why half-marathon pace differs -> current training plan.

Answer support: 10 km in 53:32 gives 5:21.2/km; printed 21 km in 2:17:01 gives 6:31.5/km. Position 287 has no supplied denominator. Reasons for performance differences are personal.

Sources: [CERT]

Mountaineering to judgement

Why NIMAS -> actual course dates -> A grade -> highest altitude actually reached -> equipment purpose -> acclimatisation -> warning signs -> turn-back decision -> advanced-course plan.

Answer support: recommendation is not advanced-course completion; no summit altitude is inferred. Medical facts are educational; use qualified supervision.

Sources: [CERT], [NIMAS], [ALT]

NSS to civic action

Motto -> service hours -> adopted community -> weekly camp role -> output versus outcome -> privacy -> one follow-up -> what you would improve.

Answer support: Not Me But You; 240 hours over two years; seven-day camp. Your actual role and impact need personal evidence.

Sources: [NSS], [CERT]

Remote sensing to autonomous farming

IIRS course -> remote sensing/GIS/GNSS -> NDVI -> ground truth -> field boundary -> geodetic units -> planner -> controller -> localisation failure.

Answer support: NDVI is an indicator, not a unique diagnosis; imagery is not automatically a safe machine boundary; GNSS augmentation and integrity are separate concerns.

Sources: [IIRS], [NDVI], [NAV], [GNSS]

Employer to current economy

EKL brands -> two continuing segments -> railway divestment -> revenue versus profit -> farmer financing -> RBI decision -> oil/weather risk -> affordable automation.

Answer support: use FY2025-26 company figures; 7 October repo 5.50%; forecast and actual are distinct. Do not infer your individual contribution from company results.

Sources: [EKL], [RBI], [FAO]

Certificate accuracy challenge

Which certificates are participation -> which are graded -> which are prizes -> which support current employment -> which have unclear handwriting -> which need chronology clarification.

Answer support: SIH certificates establish internal participation; IIRS merit grade B differs from attendance 73%; current joining date is not on the training certificate.

Sources: [PIQ], [CERT]

22 | Rapid-fire practice sets

Family and routine

Give your family responsibilities, monthly budget, commute, recent help at home, one disagreement you handled, daily routine and one habit you are improving.

Preparation: actual amounts and dates; do not speculate about family members' feelings or compare their abilities.

Education

Give school name, board, medium, marks, 2021-22 activity, engineering admission reason, favourite subject, final-year project and semester-V distinction.

Preparation: reconcile the programme chronology and separate overall marks from individual-semester recognition.

Employment

Give joining date, designation, team purpose, your module, one algorithm, one failure case, validation method, confidentiality limit and handover plan.

Preparation: joining date and current module details are not established by the internship certificate.

Certificates

Explain Virtual Labs, placement coordination, IARI, ROS internship, two hackathons, IPR, fNIRS assistance and the three IIRS programmes.

Preparation: distinguish participation, assistance, graded completion, prize and employment evidence.

Sports and adventure

Give course dates, actual highest altitude, climbing level, two race categories/times, volleyball role, cricket role and current training routine.

Preparation: use printed distances; no rank percentile or summit claim without evidence.

Geography and defence

Explain Agra's history/economy, cantonment, ADRDE, PTS, Mathura corps, Faridabad and Prayagraj.

Preparation: distinguish city/district/cantonment; dated sources for command assignments and officeholders.

Technical whiteboard

Explain bicycle model, turning radius, graph, A*, coverage union, field capacity, NDVI, RTK and ROS interfaces.

Preparation: state units, assumptions and limits; use hypothetical examples only when clearly labelled.

Current affairs

Explain one defence contract, HAPS trial, budget, repo decision, growth forecast, BRICS, India-China dialogue and Asian Games result.

Preparation: say the source date, distinguish event from opinion and explain why it matters to your profile.

23 | Personal worksheets and last revision

One-page life chronology

School completion -> 2021-22 activity -> B.Tech admission -> Virtual Labs periods -> IARI -> EKL training -> fNIRS assistance -> graduation -> BMC -> climbing -> current employment -> races -> current SSB entry. Fill exact months and any overlaps from records.

One-page project sheet

Problem; intended user; your contribution; input/output; mathematical model; assumptions; constraints; alternatives; test cases; measured result; failure; limitation; next step; what is confidential.

One-page hometown sheet

Sarendhi administrative unit; population with source/year; local crops and irrigation; roads; nearest services; two local problems; one feasible improvement; Agra history/economy/defence; Mathura corps; route to selection centre.

One-page certificate sheet

For every PDF page: event/course; date; organiser; level; role; result; one learning; one limitation; supporting record. Use the inventory in section 1.

One-page current-affairs sheet

Event date; verified fact; source; significance for India; connection to your profile; competing considerations; your view; strongest counterargument; what needs updating.

Priority order

First reconcile records and entry instructions. Then prepare current job, Army motivation, education chronology and family responsibilities. Next deepen hometown/defence, major internships and mountaineering. Finally practise sport rules, short courses and current-affairs chains. This is a suggested study order, not a prediction of question probability.

Do not memorise these as claims: a 6,500 m summit; national SIH finalist; continuous full-time Virtual Labs employment; 73% IIRS exam marks; a completed advanced mountaineering course; a full marathon at the BSF event; a current residence inferred from your employer; a guaranteed arm/service; a guaranteed job after SSB recommendation.

Specific evidence gaps retained: exact entry notification and terms; current Sarendhi administrative/constituency facts; personal joining date; highest altitude and routes; final-year project; DIY project details; exact quiz/essay handwriting where unclear; current local officeholders; detailed 1962/Kargil battle accounts; a complete global-news digest. This handbook is broad and profile-led, not a claim that these gaps or every possible question have been resolved.

24 | Source register and verification notes

Personal facts come from the attached documents and your current-role statement. External facts use the sources below. Source keys appear directly under the relevant questions. Primary sources are preferred; secondary sources are identified where used. Mathematical worked examples are derived teaching examples, not reported experimental data.

Source conflicts preserved: Faridabad founding date differs across official pages; PTS raising/establishment dates vary across secondary accounts; old company profiles can retain the divested railway business; older Army-command material can differ from dated 2026 reporting. The handbook does not silently merge these into one claim.

Refresh before interview: chiefs and officeholders, entry terms, reporting instructions, policy rates, economic forecasts, sports results, company leadership and organisational assignments. No claim is made that this PDF will update itself.

[ADR] Government ISTI portal: ADRDE mandate

Open source

[AGE] Agra district: economy

Open source

[AGH] Agra district: history

Open source

[ALT] CDC Yellow Book 2026: high-altitude travel and illness

Open source

[ALT2] CDC: travel to high altitudes

Open source

[ARMS] Indian Army official arms and services list

Open source

[ARMS2] Bharat Rakshak: arms and services overview

Open source

[ARMY] PIB: appointment of Army Chief, 13 June 2026

Open source

[ARUN] Arunachal Pradesh official state portal

Open source

[ARUN2] Arunachal Pradesh official ILP portal: profile

Open source

[ASIAN] Olympics.com: India Asian Games 2026 medal table

Open source

[ASTAR] Amit Patel: introduction to A*

Open source

[BMC] NIM Uttarkashi: basic mountaineering course

Open source

[BRICS] Official BRICS India 2026 portal

Open source

[BRICSD] PM India: New Delhi Declaration, 12 September 2026

Open source

[BSF] BSF official history

Open source

[BUDGET] PIB defence budget backgrounder, 3 February 2026

Open source

[CANT] Agra Cantonment Board: history

Open source

[CARRIER] Indian Navy: combined carrier operations

Open source

[CDS] HQ IDS: current CDS official biography

Open source

[CENTRAL] Indian Army official Central Command page

Open source

[CERT] User-provided Certificates .pdf. Attached document, pages 1-25; page numbers refer to PDF order

[CHINA] PM India: bilateral meeting, 12 September 2026

Open source

[COMMAND] Army commands and headquarters reference

Open source

[CONTROL] Northwestern Modern Robotics: feedback control

Open source

[CORPS] PTI / Business Standard: Mathura Strike 1 Corps, 1 April 2015

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[CORPS26] RNA: Strike One command change, 2 October 2026

Open source

[CRIC] MCC: Law 6, the pitch

Open source

[CRIC2] MCC: about the Laws of Cricket and 2026 edition

Open source

[DEAL2] Akashvani: satellite and torpedo contracts, 15 October 2026

Open source

[DEI] DEI: historical perspective

Open source

[DEI2] DEI: about the institute

Open source

[DUTY] Government of Delhi: Article 51A text

Open source

[EKL] Escorts Kubota integrated annual report FY2025-26

Open source

[FAO] FAO: sustainable agricultural mechanization

Open source

[FAR] Faridabad District Court: history

Open source

[FAR2] Faridabad district portal

Open source

[FARM] SKRAU Bikaner: field capacity and drawbar power teaching notes

Open source

[FNIRS] Peer-reviewed review: fNIRS principles

Open source

[FS] UNESCO: Fatehpur Sikri

Open source

[GNSS] ESA Navipedia: GNSS augmentation

Open source

[HAPS0] MoD official HAPS flight-trial release, 6 October 2026

Open source

[HER] UP Tourism: Agra UNESCO sites

Open source

[HISTORY] PIB: published official military histories

Open source

[HMI] HMI: basic course syllabus

Open source

[IAF] Indian Air Force official portal: Chief of Air Staff

Open source

[IARI] IARI: Agricultural Engineering Division

Open source

[ICAR] ICAR agricultural engineering curriculum resources

Open source

[IIRS] ISRO: Indian Institute of Remote Sensing

Open source

[IMF] IMF World Economic Outlook, October 2026

Open source

[IP] Legal Suvidha: patent and copyright distinction, 7 August 2026

Open source

[MOD] Ministry of Defence official directory

Open source

[NAV] ISRO: satellite navigation services

Open source

[NAV2] ISRO: navigation FAQs

Open source

[NAVY] PIB: Navy Chief assumes charge, 31 May 2026

Open source

[NDVI] NASA Earthdata: NDVI

Open source

[NIMAS] NIMAS Dirang official portal

Open source

[NISAR] NRSC Bhoonidhi data portal

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[NSS] NSS official FAQs

Open source

[PIQ] User-provided PIQ Online.pdf. Attached document, pages 1-4

[PRAY] Prayagraj district: history

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[PTS] Bharat Rakshak: IAF Paratrooper Training School database

Open source

[RANK] Indian Army ranks and equivalents, reference table

Open source

[RBI] RBI MPC resolution, 7 October 2026

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[ROBOT] Northwestern Modern Robotics: nonholonomic vehicle models

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[ROPE] UIAA: dynamic rope standard 101, 2025

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[ROS] ROS 2: interfaces, topics, services and actions

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[SEL] Indian Army: overview of officer selection

Open source

[SINDOOR] PIB initial Operation Sindoor release, 7 May 2025

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[SSB] Indian Army: common SSB interview instructions

Open source

[TAJ] UNESCO: Taj Mahal

Open source

[VOL] FIVB official volleyball rules 2025-2028

Open source

[WB] World Bank India Development Update, 6 October 2026

Open source

[WEST] Akashvani: West Asia advisories, 2 March 2026

Open source

Prepared as a study resource. Personal answers are intentionally left to you. The central preparation standard is consistency between what you claim, what you can explain and what your records support.